Supplementary Material

PRISM: an auditable pipeline for post-treatment glioma surveillance, evaluated branch by branch

Draft, 2026-10-15. Every number below is copied from the result documents in `docs/results/` *and* `docs/manuscript/` *or from* `outputs/results/`*; the file is named in each section. "Grade 4" (G4) is the primary population; "all" = all gliomas. AUC = IPCW AUC(120 d), mean over 5 repeats of the per-repeat pooled out-of-fold value, with 95% patient-cluster bootstrap CI (1,000 draws), unless stated otherwise. PRISM v1 is the first version of PRISM, described in an unpublished internal report; "PRISM's initial radiomics model" and "our initial evaluation split" are its radiomics model and its seed-62 test split. "Baseline" = PRISM's core model (clinical + timing + as-of-scan treatment, ridge), the deployed configuration and the reference for every imaging comparison.*

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S1 Pre-specification register

Analysis	Status	Plan record	Decision rule / primary quantity	Outcome
Radiomics vs baseline, nested over 70 configurations, grade 4	Pre-specified primary (development)	Publication plan §3.3	ΔAUC with 95% CI	−0.032 [−0.084, +0.017]; no gain
LUMIERE locked clinical + timing + volumes, grade-4-trained	Pre-specified external model	`EXTERNAL_LUMIERE.md`	Pooled IPCW AUC	0.608 / 0.620 (ridge / boosting)
Phase-of-care ladder, post-RT R2 + I2 vs R2	Pre-specified primary; plan in module docstring, not separately committed; a smoke run printed some I0–I2 numbers before the full run; plan unchanged	`PHASE_LADDER.md` §1	ΔAUC with CI and 100-permutation p	+0.061 [−0.038, +0.156], p = 0.21; not supported
Imaging-defined (clean) label	Pre-specified; plan snapshot saved before the run	`outputs/results/analyses/clean_label/PLAN_snapshot.md`	ΔAUC ≥ 0.03 with CI excluding 0	+0.027 [−0.033, +0.082]; not met
Re-segmentation	Rule written before any result was seen	`RESEGMENTATION.md`	Detection Δ ≥ 0.05 with CI > 0, or ET larger at progression in > 65% of pairs	−0.016 [−0.057, +0.025]; 45%; not met
CFB-GBM ADC	Pre-specified; plan snapshot saved before the run	`outputs/results/analyses/cfb_adc/PLAN_snapshot.md`	Δ(d − c) ≥ 0.03 with CI excluding 0	+0.007 [−0.019, +0.035], p = 0.151; not met
Audit layer P1 (coverage) and P2 (abstention beyond case mix)	Pre-specified; plan snapshot saved before the run	`outputs/results/analyses/audit_system/PLAN_snapshot.md`	P1 coverage ≥ 0.78; P2 CI above 0	P1 0.841 [0.790, 0.885], met; P2 +0.006 [−0.023, +0.036], not met
Split variance (500 splits)	Descriptive; designed after the single-split result was known	`SPLIT_VARIANCE.md`	—	—
Positive controls, SHAP, feature conflicts, imbalance, schedule coupling, baselines, sensitivity, utility	Secondary or exploratory	respective documents	—	—

Sequence: the clean-label, re-segmentation, CFB-GBM and audit analyses were designed after the main imaging result (no added value) was known. Each had its own rule fixed before its own run. Contrasts beyond the primary ones are not corrected for multiplicity.

S2 Cohort flow

Source: `docs/manuscript/FLOW.md` (`outputs/data/mu_flow.csv`, `outputs/results/analyses/tables/lumiere_flow.csv`). Figure: `docs/manuscript/cohort_flow.png` (Figure S1).

Supplementary Table S1a. MU-Glioma-Post.

Step	Scans	Patients	Excluded
All timepoints	596	203	–
All four sequences and mask present	594	203	2
Tumour (labels 1–3) present	592	203	2
MRI day known	589	202	3
MRI day plausible (≥ -365 d, not before an earlier timepoint)	588	202	1
Progression date known for progressors	588	202	0
MRI not after recorded death	580	202	8
Before late treatment (2nd-line, immunotherapy, new treatment)	501	199	79
Before first progression (index scans)	**308**	**155**	193
Grade 4 subset	249	126	–

Supplementary Table S1b. LUMIERE.

Step	Studies	Patients	Excluded
All rated studies	616	91	–
Post-operative (week ≥ 2 after surgery)	456	81	160
Rated SD / PR / CR / Post-Op	199	70	257
Before first progression	132	61	67
Not after recorded death (index scans)	**132**	**61**	0

Censored scans (52 MU, 39 grade 4, 9 LUMIERE at 120 d) stay in the cohort and are handled by the censoring-aware labels and IPCW.

Table S2. The full cohort table with definitions is `docs/manuscript/TABLE1_COHORT.md` (`table1.csv`). Additional rows not in main Table 1: last documented follow-up after scan, median [IQR] days 275 [107–471] (all), 285 [112–478] (G4), 462 [238–721] (LUMIERE); time to progression or censoring 125 [50–237], 111 [48–208], 168 [91–287]; progression documented after the scan at any time 220 (71%), 190 (76%), 122 (92%); LUMIERE T1c field strength 1.5 T / 3 T / other or missing 81 (61%) / 44 (33%) / 7; MU field strength not distributed. WHO grade in all gliomas: grade 1, 3 (2%); grade 2, 23 (15%); grade 3, 2 (1%); "3 vs 4", 1 (1%); grade 4, 126 (81%).

S3 Outcome, censoring rules and label formulations

Source: `docs/results/FIX_LABELS.md`, `docs/results/C1_endpoint.md`, `docs/results/C1_full_tables.md`.

Estimand. Documented first progression within H = 120 d of the index scan (90 / 180 d as sensitivities). Death without documented progression censors at the death day. Label 0 requires progression after H or progression-free follow-up ≥ H. Composite (progression or death) is a sensitivity analysis only.

Supplementary Table S3. Rules adopted in the label audit.

#	Problem found	Rule
1	Death without documented progression treated as a verified negative at every horizon	Death censors at the death day (primary); composite event as sensitivity
1b	MRIs dated after the recorded death (15/96 MU deaths)	If more than one dated record follows the death day, the death date is ignored (7 patients, 3 with cohort scans); if exactly one, that scan is excluded
2	Progression flag 0 with a progression date (3 patients); two date fields disagree (29 patients)	Any patient with a progression date is a progressor; primary date `date_of_first_progression`; reverse order as sensitivity
3	One MRI day of −965; progressor who died before progression	MRI day ≥ −365 and not before an earlier timepoint; death-before-progression flagged
4	Hazard formulation counted scans censored mid-interval as surviving the interval	Censored scans contribute only fully observed intervals
5	Follow-up extended by non-imaging dates	Kept as primary; imaging-only follow-up as sensitivity
6	Unknown RT end coded as "not early post-RT"	`early_post_rt` is 1 / 0 / missing (94 scans missing, 65 grade 4)

Effect: six scans left the cohort (314 → 308; grade 4 253 → 249). At 120 d, 17 negatives became censored (16 deaths within 120 d without progression, 1 ignored death date) and 1 negative became positive; no positive changed.

Supplementary Table S4. Label counts (positive / verified negative / censored).

Population	Label	H = 90	H = 120	H = 180
All (308 / 155)	primary	65 / 195 / 48	100 / 156 / 52	144 / 105 / 59
All	composite	79 / 195 / 34	116 / 156 / 36	161 / 105 / 42
All	alternative progression date	63 / 191 / 54	99 / 151 / 58	141 / 102 / 65
All	imaging-only follow-up	65 / 187 / 56	100 / 149 / 59	144 / 99 / 65
Grade 4 (249 / 126)	primary	62 / 152 / 35	93 / 117 / 39	131 / 75 / 43
Grade 4	composite	74 / 152 / 23	107 / 117 / 25	146 / 75 / 28
LUMIERE (132 / 61)	primary	24 / 101 / 7	47 / 76 / 9	62 / 61 / 9
LUMIERE	composite	26 / 101 / 5	51 / 76 / 5	66 / 61 / 5

LUMIERE gives a survival time without a death indicator; a recorded survival time is read as a death and used only as a censoring time in the primary label.

IPCW AUC. Cumulative/dynamic AUC at 120 d with Kaplan–Meier censoring weights (Uno et al. 2007; Blanche et al. 2013). Cases are scans with an event by 120 d, weighted by the inverse of the censoring survival at their event time; controls are scans event-free beyond 120 d. The vectorised

implementation matches scikit-survival to 2×10^{-16} (`FIX_EVALUATION.md` §1). In the bootstrap, censoring weights are re-estimated in each resample; if a resample has no censoring, $G = 1$.

Supplementary Table S5. Label formulations, grade 4, H = 120 d (`C1_endpoint.md`; ΔAUC vs naive).

Features	Formulation	AUC [95% CI]	ΔAUC vs naive [95% CI]	O/E	Slope
volumes + clinical	naive	0.651 [0.567, 0.727]	—	1.14	0.95
	verified	0.657 [0.577, 0.732]	+0.006 [−0.020, +0.030]	0.95	0.91
	IPCW	0.656 [0.578, 0.731]	+0.005 [−0.021, +0.029]	0.97	0.92
	hazard	0.653 [0.574, 0.726]	+0.002 [−0.018, +0.022]	0.87	0.50
legacy radiomics + clinical	naive	0.648 [0.553, 0.736]	—	1.13	0.52
	verified	0.655 [0.575, 0.730]	+0.007 [−0.021, +0.038]	0.96	0.51
	IPCW	0.654 [0.575, 0.730]	+0.006 [−0.022, +0.036]	0.98	0.51
	hazard	0.669 [0.583, 0.746]	+0.021 [−0.027, +0.068]	0.90	0.33

All 36 ΔAUC CIs (2 populations × 2 feature sets × 3 horizons × 3 formulations) include 0. The naive label under-predicts risk in all 12 settings (O/E 1.12–1.19 against 0.95–0.99 for verified / IPCW). The legacy radiomics pipeline in C1 predates the clipping fix (S5).

S4 Predictor definitions and the as-of-scan treatment rule

Source: `prism/features/tabular.py`; `docs/results/FIX_TREATMENT_ASOF.md`; `docs/results/FIX_FEATURES.md`; audit record input tables (`outputs/results/analyses/audit_system/records.md`).

Supplementary Table S6. Baseline predictors (23 raw; 31 design columns after missingness indicators).

Group	Predictor	Definition (as of scan day)	Source column(s)
Clinical	age	age at diagnosis, years	`clinical_age_at_diagnosis`
	male	sex at birth	`clinical_sex_at_birth`
	idh_mutant, idh_unknown	IDH1/IDH2 mutant; neither documented	`clinical_idh1_mutation`, `clinical_idh2_mutation`
	mgmt_methylated, mgmt_unknown	MGMT promoter methylated; unknown code	`clinical_mgmt_methylation`
Timing	days_from_diagnosis	scan day	`days_from_diagnosis_to_mri`
	days_since_rt_end	days since documented RT end; missing until RT completed	RT end day vs scan day
	rt_date_unknown	no completed RT with known end date at the scan	RT end day
	early_post_rt	0–84 d after RT end (RANO 12-week window); missing if RT end unknown	RT end day
	early_post_rt_x_mgmt_methylated	interaction	RT end day × MGMT
	visit_index	rank of the scan among the patient's MRIs	MRI days
Treatment	rt_started	RT start day ≤ scan day	`clinical_radiation_therapy` + RT start day
	rt_completed	RT end day ≤ scan day	RT end day
	rt_dose_gy, rt_fractions, rt_hypofractionated	prescription, visible once RT started; hypofractionation ≤ 15 fractions	`clinical_dose`, `clinical_number_of_fractions`
	rt_start_day	days from diagnosis to RT start, visible once started	RT start day
	rt_duration_days	RT end − start, visible once completed	RT start and end days
	chemo_started	initial TMZ / lomustine start ≤ scan day	initial chemotherapy fields
	surgery_day, biopsy_before_resection	visible once the first surgery has happened	first surgery fields
	previous_brain_tumor	history, known at any scan	`clinical_previous_brain_tumor`

Unknown values are imputed in-fold with the training median plus a missingness indicator; nothing is zero-filled silently. Therapy marked given without a start date (2 RT, 2 chemotherapy patients) is assumed to have started before the scan.

Supplementary Table S7. What leaked before the as-of-scan fix (scans affected, grade 4 / all gliomas).

Feature	Problem	Affected
`rt_given`, `chemo_given`	"will receive RT / chemotherapy", true for scans before the course started	13 / 23 (RT), 13 / 21 (chemo)
dose, fractions, hypofractionation, `rt_start_day`	prescription and start date of a course that had not started	13 / 19–23
`rt_duration_days`	duration of a course not yet finished (also reveals early stopping)	17 / 29
`days_since_rt_end`, `rt_date_unknown`	negative values from the future RT end date	17 / 29
`surgery_day`, `biopsy_before_resection`	one scan precedes the first surgery	1 / 1

Progression, death, last follow-up and late-treatment dates are used only for labels, censoring and eligibility. Change features use earlier scans only.

Effect on repeated CV (intermediate inputs; earlier repeat-averaged estimator; `FIX_TREATMENT_ASOF.md` §3). Baseline ridge grade 4 0.694 [0.613, 0.775] → 0.729 [0.648, 0.811]; all gliomas 0.719 [0.648, 0.788] → 0.749 [0.675, 0.820]. Boosting grade 4 0.723 [0.633, 0.806] → 0.731 [0.641, 0.813]; all gliomas 0.743 [0.666, 0.816] → 0.740 [0.663, 0.815]. Timing-only boosting fell (grade 4 0.718 → 0.696; all 0.735 → 0.705): part of its earlier signal came from the future RT end date. The MGMT × window flag changed repeated-CV AUC by ≤ 0.008. These intermediate values are superseded by the final run (main Table 2).

Molecular look-ahead found by the audit (exploratory). IDH and MGMT are used at scans taken before the diagnosis date: 8 grade 4 scans (7 patients; 1 event, 4 non-events, 3 censored) and 11 all-glioma scans, 1–58 days before diagnosis (`AUDIT_SYSTEM.md` §2). Its effect on the baseline AUC was not quantified.

Imaging measurements. Volumes (ET, NETC, SNFH, RC, tumour = ET + NETC, lesion), morphology (ET fraction, multifocality, largest axial ET area, ET-to-cavity distance and proximity, T1c / FLAIR intensities relative to brain > 10 mm (Euclidean) from lesion and cavity), and past-only change (from previous scan and nadir, growth rate, `vol_pd_flag` requiring a measurable nadir ≥ 0.1 cc, `et_new_from_zero_nadir`, missingness flags for undefined relative change; 25 change columns). For LUMIERE transport, necrosis and cavity were merged (`log_necrosis_cavity_cc`) because DeepBraTumIA labels the cavity as necrosis (`FIX_FEATURES.md` §1).

S5 Radiomics: extraction, configurations and outlier clipping

Source: `docs/results/C3_radiomics.md`, `docs/results/FIX_RADIOMICS_CLIP.md`, `docs/results/FIX_FEATURES.md` §6, `prism/features/radiomics.py`.

Extraction. PyRadiomics (van Griethuysen et al. 2017). *Legacy* features are those of PRISM's initial radiomics model: T1c and FLAIR, original, LoG and wavelet images, 2,446 columns. Under the legacy within-lesion z-score and fixed bin width the texture matrices are effectively two-level (GLCM joint entropy ≤ 2 bits), so legacy texture behaves as a binary habitat descriptor (above vs below lesion mean). *Corrected* (v2) features use a whole-brain z-score, a fixed bin count of 32 (joint entropy 4.7–6.3 bits), no second N4 correction, and are extracted per compartment (ET, NETC, SNFH, core, lesion, 5 mm peri-cavity rim, shape) and for all four sequences (2,316 columns). Known limitations: a fixed bin count rather than the IBSI-recommended fixed bin width (Zwanenburg et al. 2020); the ICC stability pass covers T1c and FLAIR texture and first-order features only (1,116 of 2,316 columns); the 5 mm rim uses a city-block dilation.

Configurations (70). Variant (legacy; corrected whole lesion; corrected per compartment; corrected all four sequences) $\times$ stability filter (none; ICC ≥ 0.75 under 1-voxel erosion, in-fold) $\times$ selection (top-10 or top-30 univariate F; L1 with non-zero coefficients only, at most 30; PCA-10; decorrelation + top-20) $\times$ integration (concatenation with the baseline; or a stacked radiomics score from inner out-of-fold predictions). Nested choice: inner 3-fold patient-grouped AUC in each outer training fold. Missing compartments are imputed with the median and one indicator per missingness pattern.

Outlier clipping. Three legacy columns (FLAIR LoG $\sigma = 1, 2, 3$ mm NGTDM Coarseness) held the PyRadiomics sentinel 10^6 in four scans (two grade 4); other values in these columns have medians of 10^{-4} to 8×10^{-4}. One such value inflates a training SD by orders of magnitude and makes PCA and the stacked score depend on these scans. Before the fix, the stacked score had inner-OOF training values of -10^4 to -4.6×10^6 in 20/25 grade 4 fits. The fix, `QuantileClip(q = 0.01)`, is fitted in each training (and inner) fold, clips to the fold's 1st/99th order statistics, and applies the same bounds to test scans. It does not look at labels and applies to all configurations and to deep features.

Supplementary Table S8. Effect of the clipping fix (`FIX_RADIOMICS_CLIP.md`).

Population	Nested AUC before → after	ΔAUC before → after	Cal. slope
Grade 4	0.672 [0.597, 0.743] → 0.666 [0.586, 0.739]	−0.026 [−0.073, +0.017] → −0.032 [−0.084, +0.017]	0.50 → 0.60
All gliomas	0.732 [0.664, 0.794] → 0.747 [0.678, 0.811]	+0.012 [−0.029, +0.050] → +0.028 [−0.017, +0.071]	0.77 → 0.85

Largest change in any single configuration: 0.020. Single configurations with CI above 0: grade 4 0/70 before and after (below 0: 12/70 → 8/70); all gliomas 0/70 → 1/70 (`legacy|none|pca10|stack|ridge`, 0.754, +0.035 [+0.002, +0.068], $p(\Delta \leq 0) = 0.021$). Not covered by the fix: the radiomics group of the feature-group search (S10), the C1 radiomics feature set (S3), and the feature-conflict B + radiomics row (S12).

Supplementary Table S9. Radiomics results by population (`C3_radiomics.md`; `outputs/results/c3/{grade4,all_glioma}/results.csv`).

Model	Grade 4 AUC	Δ [95% CI]	Slope	All gliomas AUC	Δ [95% CI]	Slope
Baseline ridge	0.698 [0.615, 0.774]	—	0.89	0.720 [0.638, 0.792]	—	0.88
Baseline boosting	0.687 [0.601, 0.768]	−0.012 [−0.072, +0.048]	0.82	0.718 [0.642, 0.792]	−0.002 [−0.052, +0.047]	0.87
Best post hoc (legacy PCA-10 stack)	0.721 [0.638, 0.793]	+0.023 [−0.013, +0.057]	0.92	0.754 [0.685, 0.823]	+0.035 [+0.002, +0.068]	0.96
Nested over 70 (honest)	**0.666 [0.586, 0.739]**	**−0.032 [−0.084, +0.017]**	0.60	**0.747 [0.678, 0.811]**	**+0.028 [−0.017, +0.071]**	0.85

Median ΔAUC by variant, grade 4: legacy +0.005, v2 all four sequences −0.016, v2 compartments −0.017, v2 lesion −0.001; all gliomas +0.024, +0.000, −0.005, +0.005. Nested choices: legacy in 18/25 (grade 4) and 25/25 (all) outer folds. Stacking beat concatenation in grade 4 (median Δ −0.003 vs −0.027).

S6 Frozen deep features

Source: `docs/results/DEEP_FEATURES.md`; `outputs/results/analyses/deep/`.

Encoders. MONAI SwinUNETR (four channels FLAIR / T1c / T1 / T2, feature size 48), encoder only: BraTS 2021 weights; BrainSegFounder (UK Biobank self-supervised pretraining, BraTS fine-tuning; Cox et al. 2024 [CHECK]); and random weights (seed 0) as the control. **Input.** Skull-stripped 1 mm volumes, z-scored within the brain, reoriented to LPS, zero-padded to a multiple of 32; eight flip test-time augmentations. **Features.** Mask-weighted mean pooling at stages 0–3 (48 + 96 + 192 + 384 = 720 per region); MU regions lesion, ET, NETC, SNFH, 5 mm rim and ET ∪ SNFH; LUMIERE regions ET, SNFH, ET ∪ SNFH and lesion. The 768-d bottleneck was not used (between-scan SD 3×10^{-4} for BraTS 2021 weights). **Families.** D1 pretrained, 32 configurations (2 encoders × {lesion, compartments} × 4 selections × 2 integrations); D0 random, 16; D1b deep + legacy radiomics stacked, 16; DS shared regions, 16. **Licences.** BraTS 2021 SwinUNETR weights Apache-2.0; BrainSegFounder weights under the UK Biobank material transfer agreement (not redistributed).

Supplementary Table S10. Deep features on MU.

Arm	Grade 4 AUC	Grade 4 Δ [95% CI]	All AUC	All Δ [95% CI]
Baseline	0.698 [0.615, 0.774]	—	0.720 [0.638, 0.792]	—
D1 pretrained, nested over 32	0.694	−0.004 [−0.069, +0.063]	0.731	+0.011 [−0.032, +0.056]
D0 random weights, nested over 16	0.649	−0.049 [−0.085, −0.014]	0.711	−0.009 [−0.032, +0.009]
D1b deep + legacy radiomics, nested over 16	0.709	+0.011 [−0.032, +0.054]	0.751	+0.031 [−0.003, +0.066]
DS shared regions, nested over 16	0.699	+0.001 [−0.057, +0.060]	0.725	+0.005 [−0.035, +0.046]
Baseline + volumes + D1 vs baseline + volumes	0.694 vs 0.702	−0.008 [−0.045, +0.033]	0.739 vs 0.733	+0.007 [−0.021, +0.036]

Single configurations with CI above 0: 0/80 (grade 4); 3/80 (all gliomas; +0.035 to +0.040, lower bounds +0.001 to +0.004); about 2 of 80 expected by chance. Image-only stacked score, grade 4: pretrained 0.652 (modal configuration), random 0.566; correlation with log tumour volume r = 0.58 and with the baseline score r = 0.28. Calibration slope of nested D1: 0.55 (grade 4), 0.79 (all).

S7 PRISM v1: the initial radiomics model, the initial evaluation split and the fusion components

Source: `docs/results/SHAP_PRISM_SPLIT.md`; `docs/manuscript/FIGURES.md` (Figure 1); `outputs/results/analyses/ shap_prism_split/`; `outputs/results/analyses/paper_figures/single_split{,_pairs}.csv`.

Reconstruction. The input matrix was rebuilt from the legacy feature files plus the clinical codes using PRISM v1's case table; the shipped bundle reproduced the stored test predictions on all 84 rows (maximum absolute difference 1.1×10^{-15} calibrated, 1.4×10^{-15} raw). The model is an L2 logistic regression (liblinear, C = 0.0087, balanced class weights) on 46 standardised columns (two of 48 selected columns are dropped by a correlation filter), with a monotone Platt calibrator. Exact linear SHAP reproduced the shipped importance table (Spearman 0.997, identical top 10). Modality shares: T1c 52%, FLAIR 26%, clinical codes 22% (share of mean absolute per-scan group sum).

Supplementary Table S11. Composition of the test set of our initial evaluation split (percentile among 500 design splits).

	Full cohort	Initial split test set	500 splits: mean (2.5–97.5%)	Percentile
Scans / patients	308 / 155	82 / 29	84.5 / 29.8	
Event rate, naive	0.325	0.268	0.254 (0.165–0.348)	62
Grade 4 / GBM	0.81 / 0.70	0.80 / 0.77	0.82 / 0.63	44 / 97
IDH-mutant scans	0.195	0.146	0.271 (0.131–0.403)	5
Pre-RT scans	0.373	0.439	0.353 (0.220–0.474)	89
Event rate in pre-RT scans	0.087	0.000 (0 / 36)	0.044 (0–0.138)	15
≤ 12 weeks post-RT	0.338	0.244	0.283	16
Later post-RT	0.263	0.268	0.345 (0.243–0.450)	6
AUC of the rule "scan is post-RT"	0.677	0.740	0.675 (0.575–0.772)	92

Supplementary Table S12. Pair decomposition (naive AUC) (`single_split_pairs.csv`; 22 positives, 36 pre-RT negatives, 24 other negatives; 60% of pairs involve a pre-RT negative).

Model	All pairs	Positives vs pre-RT negatives	Positives vs other negatives
PRISM initial radiomics model	0.802	0.823	0.771
Baseline ridge	0.871	0.962	0.735
Baseline GBM	0.844	0.933	0.710
Baseline + legacy radiomics	0.862	0.934	0.754
Clinical only (ridge)	0.791	0.789	0.793

On the 36 pre-RT scans the initial radiomics model's T1c features lowered the log-odds by −0.50 on average; the baseline ridge did the same through treatment (−0.38) and timing (−0.15). Our models' SHAP profiles on this split match their whole-cohort out-of-fold profiles at group level (baseline

ridge: treatment 55%, clinical 29%, timing 16% on the split; 52 / 29 / 19% out of fold); the split does not lean on unusual features.

Post-RT test scans (42 scans, 19 patients, 19 positives; IPCW AUC [95% CI], naive in brackets): initial radiomics model 0.770 [0.604, 0.907] (0.778); baseline ridge 0.718 [0.481, 0.880] (0.730); baseline GBM 0.699 [0.463, 0.846] (0.705); + legacy radiomics 0.758 [0.605, 0.886] (0.769); clinical only 0.818 [0.664, 0.924] (0.826), Δ vs baseline +0.100 [−0.015, +0.293].

Before and after the as-of-scan fix (30 test patients / 84 scans, naive / IPCW; `head_to_head_{before,after}.csv`): initial radiomics model 0.802 / 0.811 in both; baseline ridge 0.870 / 0.844 → 0.880 / 0.860; baseline boosting 0.901 / 0.881 → 0.869 / 0.855; baseline + legacy radiomics 0.869 / 0.848 → 0.871 / 0.852; baseline + corrected compartments 0.869 / 0.843 → 0.886 / 0.870. With the final labels (29 / 82): ridge 0.871 / 0.851, boosting 0.844 / 0.824.

PRISM v1 encoder and fusion components (source: `legacy/reproduction/{README,FINDINGS,AUDIT}.md`; v1's own case table, labels and scoring on the 30-patient / 84-scan test set). All v1 numbers reproduced from the code. The v1 image encoder was a small SwinUNETR-style transformer with random, untrained weights; its embeddings alone gave AUC about 0.50 in fusion (linear probe 0.55 in training CV, 0.62 on the test set). The v1 late-fusion MLP scored AUC 0.8006, below its radiomics input alone (0.8043); its checkpoint had been selected on test loss and its 0.48 threshold hand-tuned, so its reported recall and F1 gains over the radiomics model were an operating-point shift. A BraTS-pretrained SwinUNETR fine-tuned on MU (lesion-masked pooling) reached image-only test AUC 0.64–0.65, and late fusion with the radiomics score gave 0.790–0.809, with a patient-bootstrap CI of the difference spanning 0. These results motivated the frozen-encoder design with a random-weight control (S6) and the single regularised model with ablatable branches (main §2.1). The v1 report generator's RANO section was computed from the radiomics score, not from mask measurements; the rebuilt generator is described in S19.

S8 Split variance: full tables and verifier checks

Source: `docs/results/SPLIT_VARIANCE.md`; `docs/manuscript/FIGURES.md` §2 (clipped `baseline_rad`); `outputs/results/analyses/split_variance/{summary,summary_clipped,per_split_auc_clipped}.csv`.

Design. Seeds 0–499 of PRISM v1's split search gave 500 distinct test sets (82–94 test scans, mean 86.1; 10–33 positives). The search favours patients with many scans: all 157 patients of PRISM v1's case table appear, and the most frequent is in 71% of splits. Cohort, label and features are PRISM v1's (315 scans / 157 patients; naive label, equal to ours on all 314 shared scans). Rank = share of splits with strictly lower AUC than the model's own seed-62 value re-run here.

Supplementary Table S13. Naive AUC over 500 splits.

Model	Mean ± SD	2.5–97.5%	Seed 62 (rank)	≥ 0.802	Beats baseline	Mean Δ vs baseline
Initial radiomics pipeline, re-selected per split	0.720 ± 0.055	0.610–0.819	0.801 (92%)	7.2%	12%	−0.081
Initial radiomics model, shipped features refitted (optimistic)	0.815 ± 0.044	0.726–0.887	0.802 (38%)	62%	58%	+0.014
Baseline ridge	0.801 ± 0.052	0.694–0.895	0.871 (91%)	50%	–	–
Baseline + legacy radiomics (clipped)	0.799 ± 0.053	0.674–0.892	0.862 (89%)	50.2%	51.6%	−0.002

Supplementary Table S14. IPCW AUC over 475 splits.

Model	Mean ± SD	Seed 62 (rank)	Beats baseline	Mean Δ
Initial radiomics pipeline, re-selected	0.727 ± 0.058 (0.602–0.832)	0.816 (94%)	25%	−0.042
Shipped features refitted (optimistic)	0.825 ± 0.045 (0.737–0.901)	0.817 (40%)	81%	+0.055
Baseline ridge	0.769 ± 0.056 (0.654–0.876)	0.851 (92%)	–	–
Baseline + legacy radiomics (clipped)	0.774 ± 0.058	0.845 (92.0%)	56.8%	+0.005

Per split, clipped minus pre-clip naive AUC of the radiomics arm: mean −0.002, SD 0.009. Gap between honest and optimistic versions of the initial radiomics pipeline: 0.095 ± 0.040.

Composition. Baseline ridge per-split AUC correlates with the pre-RT event rate (Spearman −0.61 naive, −0.57 IPCW), the AUC of the "post-RT" rule (+0.57, +0.51), the later post-RT share (−0.26) and the pre-RT share (+0.22). Nine composition variables explain CV R^2 0.42 and predict 0.851 for seed 62 (observed 0.871, mean 0.801). For the initial radiomics pipeline: CV R^2 0.10 (naive), 0.15 (IPCW); strongest correlates IDH-mutant fraction (−0.30) and the "post-RT" rule (+0.22); predicted 0.773 for seed 62 (observed 0.801, mean 0.720).

Verifier checks (earlier per-split table; not repeated on the final run). (1) On seed 62 and 15 other splits, redoing the ranking at full cost (10 folds × 100 permutation repeats) gave 0.718 ± 0.054 on the 15 splits (cheap ranking 0.726 ± 0.057; cheap − full +0.008 ± 0.033, r = 0.83). Re-running the shipped Optuna search as well gave 0.654 ± 0.054 (−0.064 ± 0.055 against full ranking with the shipped C); its inner-CV objective averaged 0.865 (optimism ≈ 0.21), and on seed 62 it gave 0.762, not 0.802. (2) Against 300 fully random 30-patient splits, the design splits had more test scans (85.5 vs 59.9) and a narrower baseline AUC spread (SD 0.054 vs 0.069), so spread under the design underestimates what a fresh split would show (Varoquaux 2018).

S9 Estimator and evaluation details

Source: `docs/results/FIX_EVALUATION.md`.

- **Headline estimator.** Mean over the 5 repeats of the per-repeat pooled-OOF IPCW AUC (and Brier, O/E, slope). The earlier estimator — the AUC of the OOF prediction averaged over repeats — is an ensemble of five models per scan and is optimistic, more so for unstable nested pipelines. On intermediate inputs, the nested radiomics ΔAUC moved from +0.009 (repeat-averaged) to −0.002 (mean over repeats) under the change.
- **Bootstrap.** Patient-cluster bootstrap, 1,000 draws; each draw resamples patients once and recomputes the mean-over-repeats statistic of every model on that draw, so differences are paired. ΔAUC is the point estimate (not the bootstrap mean) with the bootstrap CI.
- **Calibration.** O/E (Kaplan–Meier risk / mean prediction, or IPCW-weighted), calibration-in-the-large with a logit offset, calibration slope by IPCW-weighted logistic regression on the logit, flexible calibration curves (restricted cubic spline, 3 knots) and ICI (Van Calster et al. 2019).
- **External estimator.** Every LUMIERE arm is scored on the same 5×5 patient-grouped LUMIERE folds (per-fold IPCW AUC averaged over 25 folds, censoring weights from the whole LUMIERE cohort); locked models are also scored pooled. An earlier pooled score of differently recalibrated folds produced an artefactual AUC of 0.414 for recalibrated ridge; recalibration leaves discrimination unchanged by construction.
- **Selection inside folds.** Feature selection, PCA, ICC filtering, stacking and hyperparameter choice are all inside the training fold, with inner patient-grouped CV for every choice (Varma and Simon 2006; Cawley and Talbot 2010; Demircioğlu 2021).

S10 Feature-group search with permutation control

Source: `docs/results/C2_feature_search.md`; `docs/results/FINAL_RESULTS.md`.

Six groups — clinical, timing, volumes, morphology, change, legacy radiomics (top 20 in-fold; unclipped) — give 63 combinations $\times$ 2 learners (ridge; fixed shallow boosting). The treatment block is not part of this search. The honest estimate selects one of the 126 candidates per outer fold by inner CV. The permutation control moves each patient's whole outcome block to another patient with the same number of scans, keeps features and folds, and re-runs the full nested pipeline 100 times; $p = (1 + \#\{\text{perm} \geq \text{obs}\}) / 101$.

	Grade 4	All gliomas
Post-hoc best configuration (optimistic)	boosting, clinical + timing: 0.714 [0.643, 0.785]	ridge, clinical + timing + volumes + morphology + radiomics: 0.744 [0.679, 0.804]
Nested over 126 (honest)	0.664 [0.601, 0.722], slope 0.49	0.707 [0.647, 0.766]
Permutation null mean (95th percentile)	0.548 (0.636)	0.528
Permutation p	0.050 (4/100 ≥ observed)	0.010 (0/100)

Single groups, grade 4 (ridge / boosting): radiomics 0.675 / 0.628; morphology 0.650 / 0.538; volumes 0.611 / 0.573; clinical 0.585 / 0.612; timing 0.559 / 0.649; change 0.537 / 0.565. The permutation null is centred above 0.5 because selecting the best of 126 configurations is optimistic even without signal. Boosting was chosen in 15/25 outer folds; selected boosting models had large train–test gaps (for example training AUC 0.961 for the best grade 4 configuration).

S11 Phase-of-care ladder: full results

Source: `docs/results/PHASE_LADDER.md`; `outputs/results/analyses/phase_ladder/{results, cohorts,permutation, subset_of_full,nested_choices}.csv`; `docs/manuscript/figures/fig4_phase.csv`.

Phase definitions (grade 4 / all-glioma scans): pre-RT, `rt_started == 0`, including patients with no RT documented (76 / 115); during RT, started but not completed (6 / 8); early post-RT, completed and ≤ 84 d after RT end (98 / 104); later post-RT, > 84 d (69 / 81). RT without a documented start date counts as started; RT without an end date is never "completed".

Supplementary Table S15. Cohorts.

Population · cohort	Scans	Patients	Events	Non-events	Censored	EPP R2 (10 params)	EPP R2 + I2 (54)
Grade 4 · full	249	126	93	117	39	9.3	1.7
Grade 4 · post-RT	167	84	83	83	1	8.3	1.5
Grade 4 · later post-RT	69	40	36	32	1	3.6	0.7
All · full	308	155	100	156	52	10.0	1.9
All · post-RT	185	92	84	96	5	8.4	1.6
All · later post-RT	81	46	37	40	4	3.7	0.7

By the Riley criteria (anticipated R^2cs 15% of maximum), the primary post-RT cohort would need about 750 scans (375 events) for R2 and about 4,040 scans (2,022 events) for R2 + I2. Nearly all censoring is in pre-RT and early scans. In post-RT cohorts two phase dummies are constant, so R2 has about 8 effective parameters (exploratory note).

Supplementary Table S16. Primary and key secondary results (grade 4, post-RT, ridge).

Model	AUC (95% CI)	Δ vs R2 (95% CI)	Slope	O/E
R2 (clinical + phase)	0.549 (0.458–0.642)	—	0.76	0.99
R2 + I2 (primary)	0.610 (0.529–0.688)	+0.061 (−0.038, +0.156)	0.68	0.99
R2 + I3 (key secondary)	0.609 (0.543–0.676)	+0.061 (−0.041, +0.157)	0.44	0.96
R2 + I1	0.593	+0.044 (−0.054, +0.137)	0.60	0.98
R2 + I4	0.625	+0.076 (−0.041, +0.192)	0.30	0.97
R3 (baseline), reference	0.604 (0.520–0.690)	—	0.85	0.99

Permutation test of the primary contrast: p = 0.208 (20 of 100 permutations ≥ +0.061; p = 21/101); null mean +0.028, median +0.025, range −0.104 to +0.162. The nested radiomics choice was unstable (most common configuration in 12 of 25 outer folds).

Supplementary Table S17. Bases by cohort (grade 4, ridge; AUC [95% CI]) (`fig4_phase.csv`).

Base	Full	Post-RT	Later post-RT
R0 clinical	0.585 [0.496, 0.666]	0.552 [0.456, 0.646]	0.343 [0.281, 0.418]
R1 + timing	0.644 [0.560, 0.721]	0.572 [0.485, 0.667]	0.388 [0.291, 0.503]
R2 + phase	0.664 [0.569, 0.753]	0.549 [0.458, 0.642]	0.376 [0.294, 0.470]
R3 baseline	0.698 [0.615, 0.774]	0.604 [0.520, 0.690]	0.472 [0.339, 0.617]

Full grade 4 cohort: R1 − R0 +0.059 (−0.016, +0.133); R2 − R0 +0.080 (+0.005, +0.152); R3 − R0 +0.113 (+0.026, +0.208); R3 − R2 +0.034 (−0.027, +0.090). R2 + I1 0.696 vs R3 0.698 (Δ −0.002, −0.081 to +0.074). Imaging on top of each base (full grade 4; I1 / I2 / I3 / I4): R0 +0.074 / +0.071 / +0.065 / +0.063; R1 +0.030 / +0.026 / +0.008 / +0.018; R2 +0.031 / +0.023 / +0.002 / +0.020; R3 +0.004 / −0.007 / −0.032 / −0.004; no CI excludes 0. Post-RT grade 4 (I1 / I2 / I3 / I4): R0 +0.049 / +0.058 / +0.046 / +0.079; R1 +0.030 / +0.038 / +0.042 / +0.060; R2 +0.044 / +0.061 / +0.061 / +0.076; R3 +0.008 / +0.017 / +0.005 / +0.039; no CI excludes 0. All gliomas, full: R0 0.619, R1 0.693, R2 0.700, R3 0.720. Boosting (secondary): grade 4 full R2 0.698, R1 0.693; imaging never helped boosting (−0.027 to +0.007); post-RT R2 + I2 vs R2 +0.012 (−0.069, +0.102).

Sensitivity analyses. Later post-RT only (grade 4: 69 scans, 40 patients, 36 events): bases below chance (R0 0.343, calibration slope −2.0), the known negative bias of CV for near-null models with few events; large ΔAUCs there (e.g. R2 + I3 +0.250) are measured against a below-chance reference and are not evidence. Full-cohort models evaluated on post-RT scans only: grade 4 R2 0.570, R2 + I2 0.620, Δ +0.050 (−0.061, +0.157); R3 0.638, R3 + I2 Δ −0.014. Exploratory: age alone 0.586 in grade 4 post-RT; age + I2 vs age +0.022 (−0.103, +0.145). LUMIERE could not be used: no RT dates; the rater flag marks 27 of 132 scans and leaves 83 scans with unknown phase.

S12 Secondary analyses

S12.1 Literature baselines and tuning

Source: `docs/results/BASELINES.md`.

Supplementary Table S18. MU ladder (reference = baseline ridge).

Model	Grade 4 AUC [95% CI]	Δ [95% CI]	All AUC [95% CI]	Δ [95% CI]
Baseline ridge (reference)	0.698 [0.615, 0.774]	—	0.720 [0.638, 0.792]	—
Clinical only	0.585 [0.496, 0.666]	−0.113 [−0.208, −0.026]	0.619 [0.540, 0.696]	−0.101 [−0.177, −0.031]
Timing only	0.559 [0.475, 0.637]	−0.139 [−0.214, −0.070]	0.607 [0.528, 0.684]	−0.112 [−0.175, −0.059]
Clinical + timing	0.644 [0.560, 0.721]	−0.054 [−0.103, −0.012]	0.693 [0.617, 0.766]	−0.027 [−0.061, +0.007]
B0 log ET volume	0.636 [0.541, 0.723]	−0.062 [−0.183, +0.043]	0.673 [0.591, 0.752]	−0.046 [−0.142, +0.054]
B0 ΔET vs previous	0.546 [0.479, 0.614]	−0.153 [−0.241, −0.056]	0.531 [0.465, 0.590]	−0.189 [−0.285, −0.101]
B1 volumetric RANO-style rule	0.551 [0.480, 0.620]	−0.147 [−0.239, −0.062]	0.554 [0.488, 0.615]	−0.166 [−0.258, −0.087]
B1 rule, RANO 2.0 window	0.508 [0.440, 0.577]	−0.191 [−0.289, −0.098]	0.509 [0.448, 0.568]	−0.210 [−0.301, −0.119]
B4 Hashim-style Cox	0.628 [0.538, 0.711]	−0.070 [−0.179, +0.031]	0.678 [0.596, 0.755]	−0.041 [−0.120, +0.039]

The B1 rule: ET ≥ 40% and ≥ 1 cc above the past-only nadir (measurable nadir ≥ 0.1 cc), or ≥ 1 cc from a ∼0 cc nadir; positive in 25% of grade 4 scans. Nested hyperparameter grids: every tuned arm within ±0.025 of the baseline, all CIs include 0 (e.g. nested over ridge + boosting −0.024 [−0.061, +0.011], grade 4); choices were unstable.

S12.2 Sensitivity of the primary comparison

Source: `docs/results/SENSITIVITY.md`.

Sensitivity	Scans / patients	Events / censored	Baseline AUC [95% CI]	+ radiomics (nested)	ΔAUC [95% CI]
Primary: grade 4, 120 d	249 / 126	93 / 39	0.698 [0.615, 0.774]	0.666	−0.032 [−0.084, +0.017]
Horizon 90 d	249 / 126	62 / 35	0.663 [0.577, 0.739]	0.664	+0.001 [−0.057, +0.063]
Horizon 180 d	249 / 126	131 / 43	0.667 [0.573, 0.752]	0.636	−0.032 [−0.092, +0.022]
Composite (progression or death)	249 / 126	107 / 25	0.642 [0.566, 0.713]	0.629	−0.013 [−0.073, +0.052]
Alternative progression date	244 / 124	92 / 39	0.670 [0.578, 0.757]	0.648	−0.023 [−0.066, +0.019]
Imaging-only follow-up	249 / 126	93 / 43	0.685 [0.600, 0.761]	0.667	−0.018 [−0.070, +0.033]
All gliomas	308 / 155	100 / 52	0.720 [0.638, 0.792]	0.747	+0.028 [−0.017, +0.071]
Excluding 12-week post-RT scans	151 / 90	46 / 39	0.725 [0.602, 0.834]	0.727	+0.002 [−0.064, +0.067]
Excluding patient 0145 (conflicting dates)	247 / 125	93 / 39	0.691 [0.602, 0.776]	0.687	−0.004 [−0.058, +0.048]

S12.3 Class imbalance

Source: `docs/results/IMBALANCE.md`. Event shares are 44% (grade 4) and 39% (all gliomas), a mild imbalance. Over learners (ridge, boosting, XGBoost, LightGBM, nested variants) × re-balancing (none, class weights, patient-aware oversampling, SMOTE with neighbours from other patients), ΔAUC against the ridge baseline ranged from −0.031 to +0.023; 4 of 190 CIs excluded 0 (ridge, all gliomas, +0.009 to +0.014). Re-balancing shifted predicted risk upward as expected from a prior shift (median O/E 1.01 without re-balancing vs 0.88–0.89 with it) and raised Brier. Grade 4 ridge with class weights: ΔAUC +0.008 [−0.001, +0.019], O/E 1.04 → 0.94. Fixed small-data XGBoost: +0.006 [−0.060, +0.072] (grade 4); nested-tuned XGBoost had calibration slope about 0.6 and transported worse to LUMIERE (0.562 vs 0.608). No re-balancing is used in any reported model.

S12.4 Schedule coupling

Source: `docs/results/SCHEDULE_COUPLING.md`. The interval to the next MRI is unknown at prediction time and partly caused by the outcome (scans are brought forward when progression is suspected), so conditioning on it is a conservative test. Grade 4: positive rate 0.54 when the next scan is ≤ 60 d away and 0.10 when > 180 d; interval alone AUC 0.639 [0.555, 0.721]. Baseline AUC within interval tertiles 0.718 / 0.762 / 0.617; conditional AUC (pairs within a tertile) 0.687 [0.595, 0.768]; logit coefficient per SD 0.91 → 0.86 [0.43, 1.41] after adjusting for log interval; baseline features cannot predict the interval (grouped-CV R^2 −0.05). Excluding early post-RT scans and scans whose next scan is > 120 d away: 0.777 [0.653, 0.891] (85 scans). Nested radiomics conditional AUC 0.654, below the baseline. In LUMIERE the interval alone scores 0.788 [0.703, 0.868], and the + volumes boosting model's coefficient is no longer distinguishable from 0 after adjustment (0.37 [−0.14, 0.89]).

S12.5 Feature conflicts

Source: `docs/results/FEATURE_CONFLICTS.md`. No feature group measurably hurts: every leave-one-group-out contrast that would mean "group hurts" has a CI including 0; adding any imaging group to the baseline changes AUC by −0.004 to +0.019 (CIs span 0). The conflict is redundancy: timing and treatment encode the same RT-status information (interaction −0.066 to −0.073, CI excluding 0 in 3 of 4 population × learner cells). Within the baseline, 9 design columns are exactly collinear (VIF = ∞) and 16 have VIF > 10; "RT completed" appears in at least five columns. Pre-stated pruning evaluated nested: ΔAUC −0.018 to +0.011, all CIs include 0; the same rule applied once globally gives +0.009 to +0.018, which is selection optimism.

S12.6 SHAP attributions

Source: `docs/results/SHAP.md`. Ridge: exact linear SHAP on in-fold standardised features; boosting: TreeSHAP (path-dependent), additivity checked to 10^{-6}; explanations out of fold, 25 fits. Group share = mean |per-scan group sum| / sum over groups.

Model (grade 4)	OOF AUC	Group shares
Baseline ridge	0.698	treatment 58%, clinical 25%, timing 17%
Baseline boosting	0.687	treatment 51%, clinical 35%, timing 14%
Full non-radiomics ridge	0.692	treatment 34%, morphology 21%, change 15%, clinical 13%, volumes 10%, timing 8%
Full non-radiomics boosting	0.694	treatment 26%, morphology 23%, clinical 21%, change 13%, volumes 13%, timing 3%
Baseline + radiomics score (post hoc configuration)	0.721	treatment 48%, clinical 20%, timing 16%, radiomics 16%

Time-relative features hold 35% (ridge) / 47% (boosting) of feature-level credit in the baseline. Imaging takes 45% / 51% of credit in the full models without an AUC gain. On LUMIERE, the locked ridge's under-prediction is driven by `idh_unknown` (mean −0.71 logits; mean |SHAP| 0.77 on LUMIERE vs 0.13 on MU), by lower DeepBraTumIA ET volumes (−0.32 logits) and by a missingness indicator that is never set in LUMIERE.

S12.7 Calibration, decision curves, added value and sample size

Source: `docs/results/CLINICAL_UTILITY.md`.

Model (grade 4)	AUC	O/E (weighted)	CITL [95% CI]	Slope [95% CI]	ICI
Baseline ridge	0.698	0.96	−0.09 [−0.42, 0.28]	0.89 [0.50, 1.43]	0.032
Baseline boosting	0.687	0.94	−0.12 [−0.47, 0.23]	0.82 [0.44, 1.31]	0.039
Baseline + radiomics (nested)	0.666	0.94	−0.13 [−0.46, 0.25]	0.60 [0.33, 0.93]	0.063
Baseline + deep (nested D1)	0.694	0.94	−0.15 [−0.52, 0.28]	0.55 [0.28, 0.91]	0.085

Net benefit (grade 4, IPCW)	10%	20%	30%
Treat all	0.369	0.291	0.189
Baseline ridge	0.372	0.300	0.232
Δ radiomics (nested)	−0.000 [−0.004, 0.004]	−0.001 [−0.013, 0.011]	−0.020 [−0.038, −0.004]
Δ boosting	−0.002 [−0.007, 0.001]	−0.007 [−0.021, 0.006]	−0.011 [−0.039, 0.013]
Δ deep (nested D1)	−0.002 [−0.013, 0.006]	+0.002 [−0.014, 0.018]	+0.001 [−0.023, 0.026]

Added value vs baseline (continuous NRI; IDI): radiomics +0.17 [−0.05, 0.41]; +0.000 [−0.030, 0.029]. Deep +0.48 [0.24, 0.73]; +0.036 [−0.009, 0.082]. The positive NRI reflects upward risk shifts in events by over-fitted models; ΔAUC and Δ net benefit are null (Pencina et al. 2008). All gliomas: radiomics Δ net benefit +0.011 [0.004, 0.020] at 10% and +0.014 [0.001, 0.027] at 20%, with ΔAUC +0.028 [−0.017, +0.071]; nothing similar in grade 4. LUMIERE locked boosting: net benefit ≈ treat-all (0.218 vs 0.222 at 20%).

Sample size (Riley et al. 2020; grade 4). 210 verified scans, 93 events, outcome proportion 0.443, R^2cs 0.119 (from AUC 0.698). Baseline with 31 design columns: 3.0 events per parameter; required 2,190 scans (970 events) by the shrinkage criterion. Adding 10 radiomics features: 2,896; the full radiomics pool (2,477 candidate parameters): 174,942. Requirements assume independent scans, so the shortfall is understated.

S13 External evaluation on LUMIERE: full results

Source: `docs/results/EXTERNAL_LUMIERE.md`; `outputs/results/external_lumiere/results.csv`.

Shared features: age, sex, IDH and MGMT (with unknown), log days since surgery, < 3 months post-RT flag, log ET / necrosis + cavity / SNFH volumes (DeepBraTumIA; 14 of 132 index scans without usable volumes, imputed with MU training medians inside the locked pipeline).

Supplementary Table S19. Models trained on MU grade 4. Pooled locked AUC [95% CI] and O/E; other columns are per-fold means.

Features	Learner	MU CV	Locked, pooled	O/E	Locked, per fold	Fine-tuned (shrink)	Fine-tuned (pooled)	LUMIERE only
Clinical	ridge	0.585	0.610 [0.487, 0.737]	1.15	0.558	0.544	0.505	0.396
Clinical	boosting	0.625	0.627 [0.514, 0.731]	0.86	0.624	—	0.560	0.520
Clinical + timing	ridge	0.612	0.610 [0.497, 0.737]	0.98	0.603	0.591	0.603	0.586
Clinical + timing	boosting	0.664	0.632 [0.501, 0.748]	0.84	0.643	—	0.632	0.627
Volumes	ridge	0.611	0.545 [0.410, 0.653]	1.00	0.521	0.531	0.567	0.569
Volumes	boosting	0.591	0.529 [0.389, 0.644]	1.00	0.515	—	0.513	0.518
Clinical + timing + volumes	**ridge**	0.667	**0.608 [0.469, 0.739]**	**1.33**	0.598	0.606	0.617	0.580
Clinical + timing + volumes	**boosting**	0.675	**0.620 [0.495, 0.733]**	**0.99**	0.608	—	0.658	0.610

Recalibration equals locked per fold (pooled O/E 0.93–0.97). No adaptation arm has a ΔAUC CI above 0; LUMIERE-only is significantly worse for clinical ridge (Δ −0.162 [−0.238, −0.066]). Cohort normalisation of volumes (rank) raised the pooled locked AUC of every volume model (+0.007 to +0.069); clinical + timing + volumes boosting 0.620 → 0.659 [0.529, 0.767], per-fold Δ +0.047 [−0.002, +0.110]. This is transductive and a sensitivity analysis only. All-glioma-trained locked clinical + timing + volumes: ridge 0.599 [0.462, 0.728] (O/E 1.30), boosting 0.619 [0.477, 0.740] (O/E 1.04).

Deep features on LUMIERE (118 scans with deep features; `deep/lumiere/locked.csv`): grade 4 clinical + timing 0.622 → + deep 0.679 (Δ +0.057 [−0.132, +0.235], O/E 2.08); random encoder same configuration 0.667 (+0.045). With volumes: 0.635 → 0.725 (+0.089 [−0.026, +0.215]). All gliomas: +0.064 [−0.011, +0.144]; with volumes +0.050 [+0.002, +0.109]. Pretrained vs random, same configuration: +0.011 to +0.069, no CI above 0. A classifier separates LUMIERE from MU on deep features with AUC 0.82 / 0.79 / 0.81 (BraTS 2021 / BrainSegFounder / random).

PRISM's initial radiomics model, as shipped (legacy features re-extracted, 118 scans; files dated 2026-10-09, before the final LUMIERE label definition): raw 0.613 [0.500, 0.729] pooled (O/E 1.24); recalibrated O/E 0.98; fine-tuned 0.570 pooled.

Literature baselines on LUMIERE (locked; `BASELINES.md`): clinical + timing 0.610 [0.497, 0.737]; Hashim-style Cox 0.640 [0.508, 0.754] (Δ +0.030 [−0.080, +0.127]); volumetric RANO-style rule 0.559.

Harmonisation note. DeepBraTumIA ET and SNFH medians (grade 4: 0.61 and 19.11 cc) are far below MU's (5.45 and 42.85 cc). The harmonised necrosis + cavity definition is correct but did not improve transfer in a development run (`FIX_FEATURES.md` §1).

S14 Imaging-defined label, imaging diagnostics and re-segmentation

Source: `docs/results/CLEAN_LABEL.md`; `docs/results/DIAGNOSTICS_IMAGING.md`; `docs/results/RESEGMENTATION.md`; `outputs/results/analyses/{clean_label,diagnostics_imaging,reseg}/`.

Imaging-defined label. MU codes the type of first progression: imaging, clinical only, or both. Censoring the clinical-only progressions at their progression date changed grade 4 events from 93 (44 / 12 / 37 by type) to 81 and censored scans from 39 to 51; no non-event changed. Primary (grade 4, full, ridge): clinical 0.563 (0.474–0.642); baseline 0.696 (0.610–0.776); imaging (volumes + change) 0.628 (0.549–0.701), Δ −0.068 (−0.167, +0.033); baseline + imaging 0.722 (0.639–0.799), Δ +0.027 (−0.033, +0.082). On the original label, baseline + imaging was 0.694 (Δ −0.004, −0.060 to +0.051). Secondary contrasts (post-RT, all gliomas, boosting) gave baseline + imaging Δ +0.008 to +0.028 with all CIs including 0; imaging alone in all gliomas was worse than the baseline (−0.090, −0.176

to −0.007). Within-LUMIERE CV with the expert RANO label: every model at or below chance (ridge clinical + timing 0.466), so LUMIERE cannot answer the label question. Rule not met.

Volume cross-check against the dataset's own volume table (pass). Pearson r: ET 0.9998 (exact voxel match 98.3%), SNFH 0.9994 (99.7%), RC 0.9993 (94.1%), NETC 0.9957 (83.7%); disagreements are concentrated in seven patients with extra timepoints or a second mask version in the table. Measurement volumes equal mask voxel counts exactly. Label mapping: diagonal agreement 0.94–0.97, off-diagonal ≤ 0.003. Intensities agree with labels (ET/brain on T1c median 1.45, > 1 in 98% of scans; SNFH/brain on FLAIR median 1.44, > 1 in 100%).

Concurrent detection (positive control; fail). Progression scan = first MRI with a mask on or after the documented progression date (120 scans, all gliomas).

Comparison (imaging only, 5 × 5 grouped CV; ridge / boosting)	All gliomas	Grade 4
Progression scans vs all index scans	0.62 [0.57, 0.67] / 0.63	0.62 / 0.63
Progression scans vs verified 120-d negatives	0.68 [0.62, 0.74] / 0.70	0.65 / 0.66
Both groups with a prior scan	0.62 / 0.61	0.60 / 0.62

Progression scan vs the same patient's previous scan (74 pairs): ET larger in 49% (Wilcoxon $p = 0.54$); lesion larger in 65% ($p = 0.033$); lesion growth from nadir larger in 79% ($n = 42$, $p = 0.0003$); ET growth from nadir 50% ($p = 0.95$). By progression type: imaging (40 pairs) ET larger in 50%; imaging + clinical (27) 52%; clinical only (7) 29%.

Why the control fails. Between consecutive scans ≤ 14 days apart ($n = 25$), the median change was 3.2-fold in ET and 1.8-fold in lesion volume, larger than over 45–90 day gaps; across 389 pairs the size of the change was unrelated to the gap (Spearman −0.01, $p = 0.83$). Some short-gap jumps straddle a re-resection; others (four patients) are implausible.

Label–date alignment (pass). The progression date coincides with an MRI for 62% of progressors (±7 d: 69%; 30 d: 73%); 22% of progressors have no MRI on or after the progression date. Positives: median time to progression 66 d (IQR 44–92); 76% are the last scan before progression. Agreement between the 120-d label and "the next scan is the progression scan": 79% ($\kappa = 0.53$).

Horizon curve (grade 4, IPCW AUC(H); imaging / clinical + phase / both). 60 d: 0.63 / 0.55 / 0.65; 90 d: 0.62 / 0.63 / 0.68; 120 d: 0.60 / 0.66 / 0.69; 180 d: 0.58 / 0.59 / 0.63; 270 d: 0.59 / 0.55 / 0.63. Imaging stays at about 0.6 at every horizon; 30-d estimates (16 events) are not interpretable.

Data issues to report to the data owners. Three scan pairs from different patients have identical masks (0018_T5 / 0032_T5, 0060_T2 / 0062_T2, 0113_T1 / 0132_T1) and four further pairs are near-identical; a few implausible within-days volume jumps; 15 of 96 recorded deaths precede a dated MRI (S3).

Re-segmentation. BraTS 2021 SwinUNETR and BrainSegFounder were run on all 594 complete

scans (sliding window, ROI 128^3, overlap 0.5; labels written in the MU convention; RC taken from the MU mask where the new map is background, because BraTS models have no cavity class and label most of the cavity as NETC, so tumour core is not comparable).

Agreement with MU masks	Dice ET median [IQR]	Dice WT	Pearson log-volume ET / WT
BraTS 2021 SwinUNETR	0.80 [0.49, 0.91]	0.85 [0.74, 0.93]	0.92 / 0.91
BrainSegFounder	0.81 [0.56, 0.91]	0.86	0.94 / 0.92

Short-gap consistency ($\leq$ 30 d, 54 pairs): median ET fold change 2.57 $\rightarrow$ 2.17 (Δ median |log ratio| -0.17 [-0.46, -0.02]); WT unchanged. Positive controls with the new masks: primary detection (progression scan vs verified negatives, ridge) 0.681 $\rightarrow$ 0.665, Δ -0.016 [-0.057, $+0.025$] (BrainSegFounder 0.677); ET larger at progression 49% $\rightarrow$ 45% (BrainSegFounder 41%). Forward 120-d task, imaging only: all gliomas 0.644 $\rightarrow$ 0.675, Δ $+0.031$ [-0.017, $+0.084$]; clinical + phase + imaging 0.712 $\rightarrow$ 0.740, Δ $+0.029$ [-0.010, $+0.069$]. An exploratory plain-AUC boosting gain ($+0.059$ [$+0.014$, $+0.107$]) was one of more than 20 comparisons, not pre-specified and not confirmed by the IPCW ridge estimate. Verdict by the pre-specified rule: the new segmentation does not help.

S15 The auditable layer: full results and an example audit record

Source: `docs/results/AUDIT_SYSTEM.md`; `outputs/results/analyses/audit_system/`. Run: 2026-10-15, `python -m prism audit --workers 24`, 114 s. Checks (`checks.csv`): CV predictions reproduce the baseline OOF predictions (max $|\Delta|$ 1.7×10^{-16}); intercept + Σ contributions = logit (max error 1.3×10^{-15}). Code changes after the plan were confined to an as-of rule for demographics, a NaN rule for one-class strata, and a shared estimator for referral subsets; none changed a model or a rule.

Locked artifact and model card. Grade 4: 249 training scans, 126 patients, 210 verified, 93 events; C = 0.1; conformal threshold q at $\alpha = 0.2$ / 0.1: 0.613 / 0.683 (all gliomas 0.529 / 0.620). Out of scope: pseudoprogression diagnosis; cohorts without treatment fields (including LUMIERE); other institutions without validation.

Supplementary Table S20. Example audit records (out-of-fold, repeat 0, grade 4; label shown for evaluation only).

Scan	Phase	Risk [95% bootstrap]	Conformal set $\alpha = 0.2$	Flags	120-d label
0201_T3	pre-RT	0.04 [0.01, 0.27]	{0}	OOD (MGMT unknown)	0
0136_T2	during RT	0.47 [0.36, 0.77]	{0}	none	1
0117_T1	early post-RT	0.50 [0.39, 0.61]	{0, 1}	(i) 26 d after RT; (iii)	1
0059_T1	early post-RT	0.48 [0.22, 0.77]	{0, 1}	(i); (ii) rt_start_day below training 1st percentile; (iii)	0
0089_T5	later post-RT	0.45 [0.34, 0.72]	{0, 1}	(ii) rt_start_day above 99th percentile; (iii)	1
0014_T4	later post-RT	0.63 [0.44, 0.86]	{1}	none	0 (set misses)

Example record (PatientID_0117_Timepoint_1, day 85 from diagnosis), abridged from `records.md`.

- Phase of care: early post-RT ($\leq$ 84 d after RT end).
- Risk of documented progression within 120 d: 0.50 (95% bootstrap interval 0.39–0.61); conformal set at $\alpha = 0.2$ {0, 1}.
- Referral: REFER — early post-RT window (scan 26 d after RT end); uncertain (the $\alpha = 0.2$ set contains both classes).
- Log-odds: intercept -0.33 + clinical $+0.22$ + timing -0.14 + treatment $+0.24 = -0.01$ (check error 5.6×10^{-17}).
- Inputs known by scan day: yes, all. Examples: `rt_started` = 1 (source event day 21 $\leq$ scan day); `rt_completed` = 1 (day 59); `days_since_rt_end` = 26; prescription 60 Gy in 30 fractions (visible from day 21); `surgery_day` -4.
- Largest terms: `early_post_rt` -0.12; `visit_index` -0.12; `previous_brain_tumor` $+0.07$; `rt_started` $+0.07$; `male` -0.07; `mgmt_unknown` $+0.07$ (log-odds).
- Evaluation only: 120-d label 1 (progression at 111 d).

The displayed risk (model fitted on the whole training fold) and the conformal set (model fitted on 70% of training patients) can disagree in direction (e.g. 0136_T2: risk 0.47, set {0}); this occurred in 1.8% / 2.9% of singleton sets (exploratory). Median width of the 95% bootstrap interval: 0.37 (IQR 0.32–0.44) in grade 4 and 0.31 in all gliomas.

Supplementary Table S21. P1, conformal coverage at $\alpha = 0.2$ (`conformal.csv`).

Population	Overall [95% CI]	Pre-RT	During RT	Early post-RT	Later post-RT	Events / non-events	Singleton rate
Grade 4	0.841 [0.790, 0.885]	0.868 (0/5 events covered)	0.567 (6 scans)	0.871	0.806	0.824 / 0.855	0.595
All gliomas	0.823 [0.776, 0.865]	0.850 (0/10)	0.550 (8 scans)	0.852	0.790	0.698 / 0.904	0.696

Per-patient coverage 0.828 / 0.796. At $\alpha = 0.1$: 0.949 [0.916, 0.975] / 0.928 [0.896, 0.955], singleton rate 0.296 / 0.422. Conformal model AUC 0.667 / 0.709 (cost of the 70/30 split). Displayed-risk AUC on singleton scans 0.732 / 0.723 (non-singletons 0.628 / 0.594).

Supplementary Table S22. P2 and referral rules (`p2.csv`, `referral.csv`, `referral_within_phase.csv`).

	Realised coverage	AUC retained [95% CI]	Composition-matched	Random removal	P2 [95% CI]
Grade 4	0.671	0.731 [0.635, 0.812]	0.725	0.698	+0.006 [−0.023, +0.036]
All gliomas	0.685	0.735 [0.654, 0.820]	0.733	0.720	+0.003 [−0.031, +0.036]

Grade 4 rule	Referred	AUC kept	AUC referred	Random removal	Composition-matched
(i) early post-RT	39.4% (98 scans, 47 events)	0.723	0.663	0.699 [0.673, 0.723]	0.723 (the rule itself)
(ii) out of distribution	29.8%	0.671	0.665	0.698	0.695
(iii) uncertain	40.5%	0.732	0.628	0.699	0.734 [0.709, 0.760]
any of (i)–(iii)	72.6%	0.735	0.687	0.698	0.753

Within-phase AUC (grade 4 / all): pre-RT 0.555 / 0.493; during RT 0.448 / 0.671; early post-RT 0.663 / 0.678; later post-RT 0.597 / 0.644. On OOD-referred scans the model over-predicts (O/E 0.70, slope 0.66).

Shortcut audit. Unknown marker share by 120-d label (progression / no progression / censored), grade 4: IDH unknown 0/93, 6/117, 5/39; MGMT unknown 6/93, 25/117, 8/39. Dropping both unknown indicators: grade 4 0.676 [0.592, 0.752], Δ −0.023 [−0.047, −0.001]; also dropping `rt_date_unknown` Δ −0.022 [−0.045, −0.001]; all gliomas Δ −0.011 [−0.031, +0.008]. Timing only 0.559, Δ −0.139 [−0.214, −0.070]; treatment only 0.701 [0.622, 0.776], Δ +0.003 [−0.050, +0.048] (all gliomas 0.710, Δ −0.009).

Supplementary Table S23. Subgroups (grade 4; † fewer than 10 events, descriptive).

Stratum	Scans / events	AUC [95% CI]	Slope	O/E [95% CI]
GBM	214 / 86	0.68 [0.58, 0.76]	0.78	1.10 [0.90, 1.31]
Non-GBM grade 4 †	35 / 7	0.72 [0.54, 0.95]	1.34	0.66 [0.30, 1.00]
MGMT methylated	88 / 34	0.64 [0.52, 0.77]	0.62	0.99
MGMT unmethylated	122 / 53	0.72 [0.61, 0.84]	1.11	1.14
IDH wild type	208 / 87	0.66 [0.56, 0.77]	0.78	1.09
Male / female	119 / 44, 130 / 49	0.67 [0.55, 0.79] / 0.72 [0.61, 0.83]	0.85 / 0.94	1.02 / 1.06
Age $\leq$ 56.7 / (56.7, 67] / $>$ 67	109 / 30, 68 / 35, 72 / 28	0.71 / 0.64 / 0.68	0.89 / 0.73 / 0.89	0.81 / 1.39 [1.07, 1.77] / 1.06
Early post-RT	98 / 47	0.66 [0.55, 0.76]	1.06	0.95
Later post-RT	69 / 36	0.60 [0.45, 0.74]	0.50	1.03

The middle age tertile is under-predicted (O/E CI excludes 1); uncorrected for 19 strata. Field strength, scanner and race could not be audited (no acquisition metadata in MU).

LUMIERE transport audit (shared-feature ridge, grade-4-trained; reproduces AUC 0.608, O/E 1.33). Largest standardised mean differences MU → LUMIERE: `idh_unknown` +0.97 (4.4% → 40.9%), `log_et_cc` −0.84, `early_post_rt` −0.72, `log_snfh_cc` −0.56. Mean contribution shift: `idh_unknown` −0.68 logits; net linear-predictor shift −0.64. Refit without the molecular unknown indicators: LUMIERE O/E 1.33 → 0.89, slope 0.38 → 0.73, AUC 0.608 → 0.614 (all gliomas O/E 1.30 → 0.96). MU-calibrated conformal coverage on LUMIERE ($\alpha = 0.2$): 0.715 (events 0.340, non-events 0.947).

Supplementary Table S24. Explanation stability (`stability_summary.csv`).

	Grade 4: 25 CV fits	Grade 4: 200 bootstraps	All: CV	All: bootstrap
Kendall τ-b, |coef| vs locked, median (IQR)	0.63 (0.56–0.67)	0.42 (0.36–0.49)	0.54 (0.48–0.60)	0.39 (0.32–0.46)
Top-5 overlap, mean (all 5 identical)	0.67 (4%)	0.52 (1.5%)	0.65 (4%)	0.52 (2%)
Minimum sign agreement over the top 5	1.00	0.92	1.00	0.97
Features with sign agreement $\geq$ 0.9	61%	55%	68%	61%

Locked top 5 (grade 4): `rt_start_day`, `previous_brain_tumor`, `idh_unknown`, `rt_started`, `biopsy_before_resection`. Group shares of mean |contribution|: treatment 51%, timing 25%, clinical 24%. On a LUMIERE refit of the shared-feature model, 9 of 11 coefficients keep the MU sign; `early_post_rt` and the necrosis + cavity volume flip.

S16 Reproducibility

Every stage is a command of one Python package (`python -m prism <stage>`), wrapped by numbered scripts in `scripts/` (`run_all.sh` with a `--to STEP` option). Shared fold assignments are

in `outputs/data/folds.csv`; every analysis writes its outputs, a `run_info.json` and a log under `outputs/`. Seeds are fixed (bootstrap, permutations, conformal splits). The audit layer writes a `MANIFEST.json` with SHA-256 hashes of inputs, code and outputs, the git state and package versions (one recorded drift: scipy 1.17.1 installed against 1.18.1 in the lock file). Pretrained weights are downloaded locally and not redistributed. Unit tests cover label rules, as-of-scan treatment features, feature fixes and the repeated-CV estimator. Commands for each analysis are listed at the end of the corresponding result document in `docs/results/`. [PENDING: public code release and tag.]

S17 Reporting checklists cross-referenced to the manuscript

Item wording: TRIPOD+AI (Collins et al. 2024), `docs/manuscript/CHECKLIST_TRIPOD_AI.md`; CLEAR (Kocak et al. 2023 [CHECK]), `docs/manuscript/CHECKLIST_CLEAR.md`. "M" = main text section; "S" = Supplement section.

TRIPOD+AI.

Item	Topic	Where reported
1	Title	Title (development + external evaluation, population, outcome)
2	Abstract	Structured abstract
3a–c	Background, intended use, inequalities	M1; intended use in Key points and S15 model card; inequalities: race not auditable (M4 Limitations, S15)
4	Objectives	M1 final paragraph
5a–b	Data sources, dates	M2.3; accrual dates: [PENDING — relative days only in both datasets; take from dataset papers]
6a–c	Setting, eligibility, treatments	M2.3–2.4, S2; as-of treatment handling M2.5–2.6, S4
7	Data preparation	M2.5, S4, S5, S14
8a–c	Outcome, assessors, blinding	M2.4, S3; MU assessor not documented, LUMIERE expert RANO; no blinding (M4 Limitations)
9a–c	Predictors	M2.5, S4–S6; MU masks from the dataset, LUMIERE automatic
10	Sample size	Fixed by public datasets; Riley calculation S12.7, M3.8
11	Missing data	In-fold imputation with indicators (M2.5, S4)
12a–g	Analysis methods	M2.1 (architecture), M2.6–2.13, S8–S9, S13 (recalibration, locked predictions)
13	Class imbalance	None used; S12.3
14	Fairness	Subgroup audit S15; limits M4
15	Model output	Probabilities, audit record and grounded report (M2.12–2.13, S15, S19); decision curves S12.7; no threshold chosen
16	Development vs evaluation differences	M2.11, S13
17	Ethics	Declarations [PENDING]
18a–f	Funding, conflicts, protocol, registration, data, code	Declarations [PENDING]; plans S1; code S16
19	Patient and public involvement	Declarations (none)
20a–c	Flow, characteristics, comparison with development data	Table 1, S2 (Figure S1, Table S2)
21	Participants and events per analysis	Tables 1, S4, S15; each result table
22	Full model specification	Locked model files and model card (S15) [PENDING: release coefficients]
23a–b	Performance with CIs, subgroups, heterogeneity	M3, Tables 2–3, S11–S15; heterogeneity only MU vs LUMIERE
24	Model updating	S13 (recalibration, fine-tuning)
25–26	Interpretation, limitations	M4
27a–c	Usability, input quality, next steps	S15 (OOD flags, as-of checks), M4

CLEAR (selected items; full table in `CHECKLIST_CLEAR.md`).

Items	Topic	Where reported
1–6	Title, abstract, keywords, background, rationale, objectives	Title, Abstract, Keywords, M1
7–11	Guidelines, ethics, sample size, study nature, eligibility	M2.2, Declarations, S12.7, M2.4, S2
13–18	Data source, overlap, split, imaging protocol, non-radiomic predictors, reference standard	M2.3, S6 (BraTS pretraining overlap risk), M2.7–2.8, S4, S3; MU protocol not distributed
19–24	Segmentation, pre-processing, resampling, discretisation, image types	M2.3, S5, S14
25–28	Extraction method, classes, number, defaults	S5
29–33	Missing data, imbalance, reliability, scaling, dimension reduction	S4, S12.3, S5 (ICC filter), S5
34–37	Algorithm, tuning, confounders, model selection	M2.8, S5, S12.4 (schedule), M2.9 (phase of care)
38–43	Testing, metrics, uncertainty, comparisons, explainability	M2.8, M2.11, S9, S12.6
44–48	Characteristics, flow, feature statistics, performance, comparison	Table 1, S2, S5, M3, Table 3
49–52	Discussion items	M4
53–58	Open science	Declarations, S16 [PENDING: release]

S18 CFB-GBM diffusion analysis

Source: `docs/results/CFB_ADC.md`; `outputs/results/analyses/cfb_adc/{results.csv, rule.json}`; dataset facts `docs/literature/DATASETS.md`.

Data. TCIA CFB-GBM clinical v03, treatment and MRI availability v02, RANO v04, PyRadiomics features v03 on the GTV; no images used. t0 is the RT-planning MRI; t1 lies a median 14 weeks after t0, about 6–8 weeks after the end of RT, inside the 12-week pseudoprogression window. t1→t2 labels: 30 progressions among 107 labelled patients. No IDH, MGMT or progression dates. The RANO label is computed from the GTV volume (RANO sizes equal the PyRadiomics GTV mesh volume, Spearman 1.00).

Design. One row per patient; outcome t1→t2 progressive disease. Feature sets (a) clinical (age, sex, WHO performance status, WHO edition, RT dose, fractions, delay to RT, weeks t0→t1); (b) + log GTV volume at t1 and change t0→t1; (c) + T1-Gd and FLAIR first-order and texture at t1 plus GTV shape; (d) + ADC first-order and texture at t1. Each radiomics block: in-fold 1% clipping, median imputation, variance filter, scaling, top-k univariate F ($k \in \{5, 10\}$); ridge with $C \in \{0.01 \ldots 3\}$; k and C by inner stratified 3-fold CV. 20 repeats of stratified 5-fold CV, identical splits for all sets; patient bootstrap CIs (1,000); 1,000 permutations of the ADC block for the primary contrast.

Supplementary Table S25. Results (AUC = mean over 20 repeats of pooled-OOF ROC AUC).

Set	Conventional cohort (107 patients, 30 events)	ADC cohort (77 patients, 19 events)
(a) clinical	0.501 (0.403–0.596)	0.565 (0.437–0.689)
(b) + GTV volume and change	0.685 (0.569–0.781)	0.713 (0.589–0.814)
(c) + conventional radiomics	0.676 (0.575–0.769)	0.691 (0.576–0.788)
(d) + ADC radiomics	0.696 (0.593–0.785)[a]	0.698 (0.577–0.800)

Contrast	Conventional cohort	ADC cohort
(b) vs (a)	+0.185 (+0.064, +0.293)	+0.148 (+0.031, +0.266)
(c) vs (b)	−0.009 (−0.067, +0.051)	−0.022 (−0.074, +0.025)
(d) vs (c)	+0.019 (−0.006, +0.045)	**+0.007 (−0.019, +0.035)** (primary; permutation p = 0.151; minimal detectable Δ 0.038)

[a] ADC block imputed for 30 patients without ADC at t1; secondary only. **Rule not met.** Clinical features are at chance (no molecular markers and no phase variation: every t1 scan is at the same post-RT stage). Volume is the only informative block and is partly circular with the volumetric label. Conventional texture adds nothing beyond volume; ADC adds nothing measurable beyond conventional radiomics. Gains larger than about 0.04 are unlikely; smaller gains cannot be excluded. Perfusion was not available.

S19 PRISM report generator: design and faithfulness evaluation

Source: `docs/results/REPORT_AGENT.md`; `outputs/results/analyses/report_agent/` (`summary_eval.csv`, `checks_eval.csv`, `by_phase.csv`, `stress.csv`, `summary_stress.csv`, `reports/eval/`). Code: `prism/report/`, `prism/analyses/report_agent.py` (`python -m prism report`).

Plan. Written after prompt and checker development on a separate development set and before any evaluation or stress report was generated; frozen 2026-10-15 with a timestamp and the sha256 of the plan and of every report-generator code file (`PLAN_snapshot.{md,timestamp,code_sha256}`). After the run only the example-page flag labels and a duplicated "Limitations:" prefix were changed (figures re-run; no check, prompt or number changed).

Report record. The audit record (S15) plus two blocks: a risk tier (tertile cut points of the fold model's predicted risk on its own training scans; descriptive, not a threshold) and past-only indicative volumes (ET and SNFH volume, change since the prior scan and the nadir, and the volumetric rule of Supplementary S12, development AUC 0.551, labelled "indicative, not RANO"; percent changes only from a baseline ≥ 1 cc). The 120-day outcome is not in the record. A flat *facts* view (field key → pre-rounded value) is the only input of the language model and the only reference of the checker. The refitted fold models reproduce the audit stage's stored risk (maximum $|\Delta|$ 9.7×10^{-17}).

Sections. Deterministic: scan context; inputs (value, unit, source column, as-of day); risk with 95% interval, tier and 80% conformal set; contributions (intercept + clinical / timing / treatment = logit, top terms; imaging "not part of this model"); referral flags with reasons; volumetric change (indicative); a standing limitations line. Language model: the summary. *Kept from PRISM v1:* the change-from-prior and nadir table (now measured volumes, past-only), the explanation section (exact additive contributions instead of SHAP on the fusion MLP), tiered confidence language, the decision-support disclaimer. *Dropped:* the fixed 0.48 alert threshold and 0.75 "high confidence" cut (hand-set on the initial split; the core model's net benefit exceeds treat-all only above about a 20% threshold); the "RANO 2.0 rule engine" and RANO–AI concordance (computed from the model's own probability, not measurements; the volumetric rule has AUC 0.55 for 120-day progression and cannot arbitrate the model); the weighted pseudoprogression confounder score (literature weights never fitted; steroid change not in the data; its main component is already a model input and a flag); the hard-coded molecular table and the printed outcome label.

Language model and arms. Arm A (grounded): `Qwen/Qwen3.8-27B`, revision `1d4bf0f2ff6012fd82039f2fa52739d0dd7c60c0` (Apache-2.0), bf16, vLLM 0.31.0 in a separate environment, tensor parallel over 4 H100, offline (no external API; data stay on the node), non-thinking mode, greedy decoding, seed 0, structured output `{"sentences": [{"text", "fields"}]}`, at most 1,024 new tokens. Frozen 10-rule system prompt: facts only; every number copied from a field; cite field keys per sentence; state risk, interval and tier; state every active flag and the clinician-review action, never describe an inactive flag as raised; no diagnosis or RANO category; no management; hedged language; volumes indicative; 4–7 sentences. Arm B (original style): the v1 `generate_llm_summary` prompt verbatim, its context blocks filled from the same record (risk in place of the fusion probability with the original 0.48 / 0.75 phrases; the volumetric rule in the "RANO 2.0" block; the original confounder score; top 3 contributions as "SHAP drivers"; a referral-flag block added so that both arms see the same facts); no citation constraint; same model and decoding; at most 500 new tokens. Arm C: arm A's prompt with `Qwen/Qwen3.5-9B`, revision `c202236235762e1c871ad0ccb60c8ee5ba337b9a`, 1 GPU.

Checker (deterministic, patterns frozen with the plan). Per claim: every cited key must exist; every number must equal a number in a cited field up to the written rounding (magnitudes of signed values and $\times 100$ for percentages allowed); a number without a citation is unsupported. Arm B has no citations; its numbers are checked against its own prompt ("vs own input") and against the record. Per report: each active flag mentioned (keyword patterns per flag); inactive flags not described as raised; tier words consistent with the record; no forbidden content (diagnosis, RANO category as outcome, management, unhedged certainty; full patterns in the source document). An unparsed report fails every check. The checker cannot see a correct number attached to the wrong concept; meaning was checked by an AI assistant's reading (below).

Sets. Development: the 59 non-grade-4 scans (two rounds; final pass A 58/59, B 51/59, C 54/59). Evaluation: all 249 grade 4 scans (126 patients), records from the out-of-fold fits of CV repeat 0, one report per scan per arm. Primary (pre-specified): arm A, share of reports with zero unsupported

claims and every active flag mentioned, Wilson 95% CI, target ≥ 0.95. Secondary: paired differences on the same scans, 2,000-draw patient-cluster bootstrap (seed 0). Stress: 30 perturbed evaluation records (10 risk moved to another tier's mid-point, 10 flag switched on, 10 flag switched off); a report passes if it passes the checks against the perturbed record and follows the change.

Supplementary Table S26. Report faithfulness on 249 grade 4 scans (`summary_eval.csv`). CI: Wilson 95%.

	A grounded, 27B	B original prompt, 27B	C grounded, 9B
Pass (zero unsupported + all flags)	236/249 (0.948) [0.913, 0.969]	171/249 (0.687) [0.627, 0.741] (vs own input)	243/249 (0.976) [0.948, 0.989]
Pass vs the record	236	0	243
Reports with an unsupported claim	13	38 (own input); 249 (record)	6
Unsupported claims per report	0.05	0.15 (own input); 2.33 (record)	0.02
Active flags omitted	0/269	48/269 (17.8%), 47 reports	0/269
Spurious flag	0	1	0
Forbidden content, any	0%	100%	2.8% (7)
diagnosis / RANO category / management / certainty	0 / 0 / 0 / 0%	51% / 41% / 99% / 40%	0.8% / 0 / 2.0% / 0.8%
Tier stated / inconsistent	100% / 0%	57% / 44% (other scheme)	100% / 0.4%
Batched time per report	0.06 s (4 H100)	0.06 s	0.08 s (1 H100)
Single-request latency, median of 10	3.6 s	–	3.1 s

Arm A. All 13 failures: the risk sentence says "within 120 days" without citing `risk.horizon_days`; the number is correct and in the record (0 numbers absent from the record). All 13 are early post-RT scans (pass by phase: pre-RT 76/76, during RT 6/6, early post-RT 85/98, later post-RT 69/69); the development set showed the same mode once. 249/249 parsed; 0 invalid citations; 0 truncated; 5.9 sentences and 480 output tokens per report. *EXPLORATORY:* accepting a number that appears verbatim in a field cited elsewhere in the report would give 249/249; this relaxation is post hoc.

A vs B (paired, 249 scans): pass +0.261 [+0.197, +0.328]; pass vs record +0.948 [+0.919, +0.973]; unsupported claims per report −0.10 [−0.15, −0.05]; flag omissions per report (keyword-based for both) −0.19 [−0.25, −0.13]; forbidden content −1.00 [−1.00, −1.00]. Arm B states a RANO verdict in 241 reports, mentions pseudoprogression in 249, recommends management in 242 (MDT / tumour board 166, short-interval MRI 172), mentions steroids (not in the record) in 84 and calls the 120-day horizon "12-month" in 41. **A vs C:** pass −0.028 [−0.062, +0.008]; forbidden −0.028 [−0.053, −0.008] (C worse). Of C's 7 forbidden hits, 1 is a real violation ("suggesting changes that may reflect treatment effects") and 6 are pattern false positives (for example "this does not confirm progression"); one C report also gives a wrong reason for a referral, a meaning error the checker cannot see. C's 6 unsupported claims are 4 invalid keys and 2 citation omissions.

Supplementary Table S27. Stress test: 30 perturbed records (`stress.csv`; pass = checks against the perturbed record and the report follows the change; checks alone: A 28, B 17, C 29).

Arm	Risk moved to another tier	Flag switched on	Flag switched off	Total
A	8/10	10/10	9/10	27/30
B	2/10	6/10	5/10	17/30
C	9/10	9/10	10/10	28/30

The three arm-A failures, read: the horizon-citation omission (the new risk and tier were stated correctly); a new interval bound that coincided with the old risk (the new risk, interval and tier were stated correctly; C fails the same record); and an invented key (`risk.tier_cut_upper_percent`) for a value that is in the record. *EXPLORATORY:* in all 30 perturbed records the grounded text followed the changed field.

Reading by an AI assistant (not a clinician). The three pre-specified examples (first scan per phase) and 12 random arm-A reports (seed 1) were read against their facts by an AI assistant (Claude); no human or clinician has read them yet: no factual error; every active flag stated with its reason. Wording issues the checker cannot see: "95% confidence interval" for a bootstrap interval (twice); a double negative for a volume below the nadir; a pre-diagnosis scan whose "all inputs known by the scan day" is "no" (the molecular look-ahead, S15) is not mentioned in the summary, only in the deterministic section; a risk just below a tier cut point that rounds to the same value as the cut. Arm B on the same examples states "stable disease" or "Progressive Disease (PD)", recommends MDT review and short-interval MRI, and mentions steroid dosing that is not in the record.

Open issues. The primary target was missed by one report; the obvious fix (require citing the horizon field, or treat constants as citable) would be tuned on the evaluation set and needs fresh records (for example another CV repeat). The checker is lexical and has false positives on negations. Tier cut points should be shown with more decimals. Re-run identity of greedy decoding under vLLM batching was not tested. Records come from one CV repeat of one cohort, and the core model has no external validation with treatment fields. No clinician evaluated usefulness, readability or trust.

S20 Phase of care and censoring

Source: `docs/results/PHASE_CENSORING.md`; `outputs/results/analyses/phase_censoring/` (file named in each table). Code: `prism/analyses/phase_censoring.py` (`python -m prism phasecens`).

Why. The phase-of-care result (main §2.5; S11) is a gain of +0.080 [+0.005, +0.152] for R2 (clinical + phase, 0.664) over R0 (clinical, 0.585) in grade 4 (`phase_ladder/rungs.csv`). A review asked whether this gain partly comes from censoring. Of the 39 censored grade 4 scans, 38 were taken before RT. The headline IPCW AUC uses one marginal Kaplan-Meier (KM) censoring curve (`prism/models/`

`metrics.py`), which assumes that censoring does not depend on phase. A look at the data, with no model, showed that all 38 censored pre-RT scans come from the 37 grade 4 patients whose RT is never documented. For these patients the RT field and the chemotherapy field are both blank, not "No". The median follow-up of the censored scans is 0 days, and 14 of these patients died without documented progression (Supplementary Tables S28 and S29).

Supplementary Table S28. Phase of care by 120-day status, grade 4 (`descriptives_phase_outcome.csv`). Five-level phase. Censored scans: follow-up shorter than 120 days with no documented progression.

Phase	Scans	Patients	Events	Non-events	Censored	Censored, follow-up ≤ 1 d	Censored deaths without progression	Verified event rate
Pre-RT, RT pending	13	13	2	11	0	0	0	0.15
Pre-RT, RT never documented	63	37	3	22	38	20	14	0.12
During RT	6	6	5	1	0	0	0	0.83
Early post-RT	98	76	47	51	0	0	0	0.48
Later post-RT	69	40	36	32	1	1	0	0.53
All	249	126	93	117	39	21	14	0.44

In all gliomas, 47 of the 52 censored scans fall in the stratum with RT never documented (92 scans: 4 events, 41 non-events, 47 censored). The other strata have 0, 0, 1 and 4 censored scans (RT pending, during RT, early post-RT, later post-RT).

Supplementary Table S29. Grade 4 patients with RT never documented and with RT documented (`descriptives_rt_documentation.csv`). IQR in square brackets.

	RT never documented	RT documented
Patients / scans	37 / 63	89 / 186
Scans per patient, median	1 [1–2]	2 [1–3]
Age, median	65 [58–78]	61 [53–69]
Male	57%	51%
IDH wildtype / mutant / unknown (patients)	32 / 2 / 3	80 / 6 / 3
MGMT methylated / unmethylated / unknown	12 / 17 / 8	33 / 45 / 11
Chemotherapy field "Yes" / blank	0 / 37	88 / 1
Hospice field "Yes"	10	15
Scan day from diagnosis, median (range)	20 (−58 to 579)	103 (−29 to 1332)
Follow-up after the scan, median (days)	36 [0–298]	317 [178–490]
Scans with follow-up ≤ 1 d	21	1
Patients who died without progression	14	0
Patients with documented progression	3	88
Scans: event / non-event / censored	3 / 22 / 38	90 / 95 / 1

The group with RT never documented is older, was scanned early (around surgery) and has no treatment record. Almost every patient in it is lost right after the scan or dies without documented progression. It looks like a group treated or followed elsewhere, or not treated, rather than a group under surveillance before RT. Its censoring is very unlikely to be independent of outcome, since

deaths dominate.

The "unknown" row of main Table 1. `early_post_rt` is missing when the RT end date is missing (`prism/data/mu.py`). The 65 grade 4 scans with this field unknown are 63 pre-RT scans of patients with no RT documented and 2 scans during RT whose RT end date is missing (all gliomas: $94 = 92 + 2$). Every one of these scans has a phase, so the phase analysis has no unknowns. Separately, 8 grade 4 scans of 7 patients (11 scans of 10 patients in all gliomas) were taken before the diagnosis date, from 58 days to 1 day before it. At these scans IDH and MGMT were not yet known (the molecular lookahead, S15).

A third lookahead in the core model. The core model's preprocessing (`c3.BASE_PRE`) adds a missingness indicator for `early_post_rt` and for its interaction with MGMT. At a pre-RT scan this indicator is 1 when RT is never documented and 0 when RT is documented later. It therefore separates "RT pending" from "RT never documented" with information from after the scan. This is a third lookahead, next to the shortcut through the unknown markers and the molecular lookahead (S15). The audit's as-of check did not flag it, because that check reads raw values, not the missingness indicators that the imputer adds.

Plan (prespecified). The plan was written and committed (`c98febe`) before any model of this analysis was fitted or scored on the real labels. The first real run saved it to `PLAN_snapshot.md`, and it was not changed afterwards. Anything added after the results were seen is labelled EXPLORATORY. Grade 4 is the primary population and all gliomas the secondary one. Folds, pipeline (`c3.BASE_PRE` + ridge, inner choice of C) and metrics are the shared ones. The refitted core model reproduces the stored C3 predictions, and the refitted R0, R1 and R2 reproduce the stored ladder predictions, all to 1.7×10^{-16} (`checks.csv`). The main elements:

- *Five-level phase:* RT pending (RT documented and started after the scan), RT never documented, during RT, early post-RT, later post-RT. "Never documented" is only known afterwards, so the five-level phase is used to stratify the analysis and to model censoring, never as a model input.
- *Censoring schemes.* `marginal_km`: one KM curve, the headline estimator. `phase5_km` (primary alternative): one KM curve per five-level phase, so that censoring may depend on phase. `phase4_km`: one KM curve per four-level phase. `logistic_ipw`: verified scans weighted by 1/P(verified) from a logistic model on phase, age, sex and days from diagnosis, with P floored at 0.1. `verified`: unweighted verified scans. `deaths_as_controls`: death without progression before 120 days counts as a non-event. Two bounds: every censored scan set to an event, or to a non-event. Every scheme reestimates its weights in each of 1,000 paired patient-cluster bootstrap draws.
- *Cohorts*, with models refitted inside each: `full`; `rt_pathway`, which excludes the patients whose RT is never documented (this needs a treatment plan field at deployment, so it is a statement of intended use, not an as-of rule); `rt_started`, scans with RT started by the scan

day (a deployable as-of rule).

- *Corrected core models*, all with the core pipeline. `core_no_unknown`: the IDH and MGMT unknown indicators dropped. C1: `core_no_unknown` with IDH and MGMT masked at the scans taken before diagnosis. C1_mean: as C1, with unknown or masked markers set to the training fold mean. C2: C1 with an as-of `early_post_rt` (1 if RT was completed by the scan and at most 84 days have passed since its end, else 0), which is never missing and so has no missingness indicator. C2 removes all three lookaheads known now.

- *P1, censoring (primary).* Two estimates of the gain R2 − R0: its value under `phase5_km`, and its value refitted in `rt_pathway`. The marginal gain is +0.080, and half of it is +0.040. The rule: the finding is *robust* if both estimates are ≥ +0.040, *partly dependent on censoring* if exactly one is, and *not supported once censoring is handled* if neither is. The two bounds are reported alongside.

- *P2, corrected headline.* The C2 IPCW AUC under `marginal_km` with CI, ΔAUC C2 − core, calibration slope and O/E. Everything else is secondary or descriptive, with no multiplicity correction.

Supplementary Table S30. P1: the phase-of-care gain R2 − R0 under each censoring scheme and cohort (`ladder_schemes.csv`). "Change vs marginal" is a difference of differences with a paired CI. Rows in bold are the two primary estimates.

Scheme / cohort	Gain, grade 4	Change vs marginal, grade 4	Gain, all gliomas
`marginal_km`, full (headline)	+0.080 [+0.019, +0.160]	–	+0.082 [+0.023, +0.152]
`phase5_km`, full	**+0.124 [+0.039, +0.232]**	+0.045 [+0.005, +0.104]	+0.121 [+0.051, +0.198]
`phase4_km`, full	+0.125 [+0.044, +0.224]	+0.046 [+0.019, +0.090]	+0.109 [+0.043, +0.183]
`logistic_ipw`, full	+0.118 [+0.022, +0.221]	+0.038 [−0.018, +0.092]	+0.120 [+0.039, +0.196]
`deaths_as_controls`, full	+0.111 [+0.044, +0.193]	+0.032 [+0.006, +0.066]	+0.108 [+0.050, +0.172]
Bound: censored = events	−0.044 [−0.107, +0.034]	−0.123 [−0.176, −0.082]	−0.015 [−0.074, +0.045]
Bound: censored = non-events	+0.167 [+0.098, +0.248]	+0.088 [+0.049, +0.130]	+0.141 [+0.081, +0.203]
`rt_pathway` refit (1 censored scan left)	**+0.057 [+0.009, +0.117]**	–	+0.018 [−0.010, +0.052]
`rt_started` refit	+0.013 [−0.027, +0.058]	–	+0.017 [−0.013, +0.047]

Verdict under the prespecified rule: robust in grade 4. Both primary estimates clear +0.040: +0.124 under censoring stratified by phase, and +0.057 after excluding the patients with RT never documented. The CIs here come from this analysis's own bootstrap draws, so the marginal CI (+0.019 to +0.160) differs slightly from the one in `phase_ladder/rungs.csv`. Three qualifications. (1) The marginal curve was conservative, not inflating. It spreads the censoring on the scan day of the group with RT never documented over every phase, which underweights the verified pre-RT scans, mostly controls at low risk. Letting censoring depend on phase raises the phase gain and every AUC (Supplementary Table S31). (2) About 30% of the gain comes from the group with RT never documented: excluding it leaves +0.057 of the +0.080. (3) The gain does not survive the extreme bound. If all 38 censored scans of that group had progressed within 120 days, the gain would be

−0.044. The descriptives make this unlikely (14 of these patients died without documented progression, and only 3 of the 25 verified scans in the group are events), but the data cannot rule it out. In all gliomas, the secondary population, the `rt_pathway` gain (+0.018) is below the threshold, so that result would read "partly dependent on censoring".

Supplementary Table S31. Ladder and corrected core model under the alternative censoring weights, grade 4, full cohort (`schemes.csv`). The last column counts only pairs of a case and a control in the same four-level phase (`marginal_km` weights).

Arm	`marginal_km`	`phase5_km` (Δ vs marginal)	`logistic_ipw`	`deaths_as_controls`	Bounds (events / non-events)	Stratified by phase
R0 clinical	0.585	0.599 (+0.014)	0.586	0.585	0.578 / 0.568	0.595
R2 + phase	0.664	0.723 (+0.059)	0.704	0.697	0.534 / 0.736	0.577
R3 core	0.698	0.749 (+0.050)	0.728	0.726	0.561 / 0.759	0.637
C2	0.678	0.734 (+0.056)	0.713	0.708	0.545 / 0.744	0.609

For the core model the change under `phase5_km` is +0.050 [+0.021, +0.081]. Stratified by phase, R2 (0.577) is no better than R0 (0.595): phase information discriminates between phases, not within them. The core model keeps some signal within phase (0.637 [0.558, 0.720]; C2 0.609 [0.529, 0.689]), mostly from treatment detail inside the early post-RT window.

Supplementary Table S32. AUC within each phase, grade 4 (`within_phase.csv`; `marginal_km`, predictions from the full cohort). A CI is shown only for strata with at least 10 verified events and 10 verified non-events; the other strata are descriptive only.

Phase (events / non-events)	R0	R2	R3 core	C2
Early post-RT (47 / 51)	0.630 [0.519, 0.729]	0.627	0.663 [0.559, 0.768]	0.630 [0.528, 0.734]
Later post-RT (36 / 32)	0.514 [0.359, 0.662]	0.480	0.597 [0.449, 0.757]	0.582 [0.425, 0.745]
Pre-RT, RT pending (2 / 11)	0.546	0.583	0.696	0.625
Pre-RT, RT never documented (3 / 22)	0.732	0.513	0.449	0.432
During RT (5 / 1)	–	–	–	–

In all gliomas the pre-RT stratum (10 events / 58 non-events) is large enough to report: core 0.493 [0.299, 0.715], C2 0.395 [0.189, 0.609].

Supplementary Table S33. P2: corrected core models C1 and C2 on MU-Glioma-Post (`corrected.csv`; `marginal_km`, full cohort). Δ: AUC minus core AUC, paired bootstrap CI.

Arm	AUC	Δ vs core	O/E	Slope
Grade 4				
Core (R3)	0.698 [0.615, 0.774]	–	1.04	0.89
`core_no_unknown`	0.676 [0.592, 0.752]	−0.023 [−0.047, −0.001]	1.05	0.86
C1	0.675 [0.591, 0.751]	−0.023 [−0.047, −0.002]	1.05	0.85
C1_mean	0.681 [0.597, 0.759]	−0.017 [−0.040, +0.003]	1.05	0.87
C2	**0.678 [0.596, 0.755]**	**−0.020 [−0.044, +0.000]**	1.04	0.85
All gliomas				
Core (R3)	0.720 [0.638, 0.792]	–	1.05	0.88
`core_no_unknown`	0.709 [0.630, 0.782]	−0.011 [−0.031, +0.008]	1.05	0.93
C1	0.709 [0.630, 0.781]	−0.011 [−0.031, +0.008]	1.05	0.92
C1_mean	0.712 [0.636, 0.784]	−0.008 [−0.027, +0.013]	1.06	0.90
C2	0.706 [0.628, 0.777]	−0.014 [−0.036, +0.005]	1.03	0.94

Grade 4 Brier score: core 0.219, C2 0.223; calibration in the large −0.09 and −0.08. Almost all of the cost of the corrections comes from dropping the unknown marker indicators (−0.023). Masking IDH and MGMT at the 8 scans taken before diagnosis changes nothing visible. Removing the lookahead to RT documentation changes the AUC by +0.003 and moves O/E closer to 1.

Supplementary Table S34. LUMIERE transport of the shared feature model, with and without the corrections (`lumiere.csv`; 1,000-draw patient bootstrap). LUMIERE has no RT dates, so the core model cannot be applied there; the shared feature model (clinical, log days since surgery, `early_post_rt`, volumes) is refitted on MU. `shared_C2` masks MU markers taken before diagnosis and sets a missing MU `early_post_rt` to 0.

	Grade 4			All gliomas		
Model	AUC	O/E	Slope	AUC	O/E	Slope
`locked_shared`	0.608 [0.469, 0.739]	1.33 [1.01, 1.77]	0.38	0.599 [0.462, 0.728]	1.30	0.38
`shared_no_unknown`	0.614 [0.462, 0.733]	0.89 [0.67, 1.15]	0.73	0.601 [0.462, 0.717]	0.96	0.89
`shared_C1`	0.618 [0.467, 0.736]	0.90 [0.68, 1.17]	0.76	0.600 [0.459, 0.716]	0.96	0.90
`shared_C2`	0.635 [0.501, 0.745]	0.99 [0.75, 1.29]	0.97	0.611 [0.480, 0.725]	1.05	1.00

In grade 4, ΔAUC `shared_C2` − `locked_shared` is +0.027 [−0.112, +0.172]. The corrections do not measurably change discrimination on LUMIERE, but they remove the miscalibration that the indicator shortcut caused.

Supplementary Table S35. Mitigations, grade 4 (`mitigations.csv`, `cohorts.csv`, `referral.csv`). (a) Models refitted inside each scope, `marginal_km`. (b) Flags applied to the predictions from the full cohort; core / C2.

(a) Restriction to an intended use Scope	Scans / patients	Censored left	Core AUC	C2 AUC	R2 − R0
Full	249 / 126	39	0.698	0.678	+0.080
`rt_pathway` (RT documented)	186 / 89	1	0.654 [0.575, 0.739]	0.625 [0.548, 0.706]	+0.057
`rt_started` (as-of rule)	173 / 86	1	0.617 [0.539, 0.703]	0.595 [0.517, 0.682]	+0.013

(b) Referral flags for scans out of scope Flag	Rate	AUC on retained scans	Gain vs random removal	Gain vs matched removal
Pre-RT as of scan (deployable)	31%	0.629 / 0.601	−0.069 / −0.077	0 by design
RT never documented	25%	0.658 / 0.634	−0.039 / −0.044	+0.013 / +0.014

Either restriction removes 38 of the 39 censored scans, so the AUC no longer depends on any censoring model. The price is 0.04 to 0.08 of AUC, because the easy pre-RT controls leave the evaluation set. In the restricted scopes C2 is below the core model: Δ −0.029 [−0.057, −0.004] in `rt_pathway` and −0.022 [−0.044, −0.003] in `rt_started`. Neither flag improves the AUC on the scans the model keeps, since both remove scans the model ranks easily. Their value is scope honesty: the displayed AUC matches the deployment population, and the model is not judged or used on patients whose outcome is mostly unobserved. "Matched removal" removes random scans with the same phase mix as the flagged ones. Only early and later post-RT have enough events for AUCs within a phase that can be interpreted (Supplementary Table S32).

Minimal detectable effects (`mde.csv`; Hanley-McNeil, 80% power, two-sided $\alpha = 0.05$, scans treated as independent, so optimistic). Grade 4 AUC against 0.5: full cohort 0.113 (93 events / 117 non-events); early post-RT 0.164; later post-RT 0.198; pre-RT 0.393, RT pending 0.645 and RT never documented 0.507 (all three descriptive only). Paired contrasts in the full cohort, from the existing bootstrap CIs: R2 − R0 0.104; arm without unknown markers − core 0.033; treatment only − core 0.070. Changes of a few hundredths, the scale a reweighting can produce, can therefore be estimated but not told apart from zero, except for contrasts as tightly paired as the shortcut arm. No AUC within a pre-RT stratum can be interpreted.

What this means for the phase-of-care claim. The claim survives and is not an artefact of the marginal censoring weights: under censoring that depends on phase the gain is larger (+0.124), and without the patients with RT never documented it is still +0.057. It needs three qualifications. About 30% of the gain comes from a group whose outcome is mostly unobserved (38 of 63 scans censored, at a median of 0 days). Arbitrary informative censoring could erase it (−0.044 under the bound where every censored scan progressed). And phase information separates phases only: once RT has started the gain is +0.013.

S21 Report generator version 2: usefulness and public reference reports

Source: `docs/results/REPORT_USEFULNESS.md`; `outputs/results/analyses/report_usefulness/` (`summary_eval.csv`, `checks_eval.csv`, `decision_eval.json`, `by_phase.csv`, `stress.csv`, `judge_{absolute,pairwise,summary}_eval.csv`, `reports/`). Code: `prism/analyses/report_usefulness.py` (`python -m prism reportuse`); new modules `prism/report/{facts_v2,prompts_v2,usefulness,style,judge,vllm_run}.py`; the version 1 (v1) code is unchanged.

No clinician has read any v1 or v2 report. The reading of v1 reports in S19 and every reading of outputs in this section were done by an AI assistant. The LLM judge below is also an AI model. "Useful" in this section means meeting our own prespecified, lexical checklist, not a judgement by a clinician.

Background. The v1 summary (S19) passed 236/249 faithfulness checks. All 13 failures were the same omission: a risk sentence that says "120 days" without citing `risk.horizon_days`.

Public reference reports (form only). We used two public corpora to set the form of the report, not its content or accuracy (`data/public_reports/`; `docs/literature/DATASETS.md`). *RadGenome-Brain MRI* (`JiayuLei/RadGenome-Brain_MRI`, commit `0348ba42`, MIT): manually written brain MRI reports (findings, findings per sequence, impression) for public BraTS 2023 cases; we used the 230 reports of the BraTS-GLI subset. *MultiCaRe* (`mauro-nievoff/MultiCaRe_Dataset`, commit `20bb8619`; CC BY 4.0, each article with its own CC licence): open-access case reports, from which we took only the MRI sentences of 181 glioblastoma or high-grade glioma cases with post-treatment content. From both we derived, with fixed lexicons (`style.py`): section structure; words, sentences and words per sentence; Flesch-Kincaid grade level (FKGL); reading time at 200 words per minute; the share of sentences with each hedging expression and each post-treatment term; and the top bigrams of change vocabulary. RadGenome-Brain reports are preoperative, and MultiCaRe sentences come from published case narratives; neither is a set of post-treatment surveillance reports. Not used: *MR-RATE* (about 98,000 real clinical brain and spine MRI reports, CC BY-NC-SA 4.0) is the most relevant corpus, but access is gated by a form with the user's name and institution, which we did not submit on the user's behalf; it is the first choice for a follow-up comparison. *TextBraTS* reports were drafted by GPT-4o and then edited by radiologists, so they are not a human style reference, and they are preoperative only. *MIMIC-CXR* and *MIMIC-IV-Note* need credentialed PhysioNet access and hold chest or general reports; not attempted.

Supplementary Table S36. Form of the public reference reports and of v1 (`reference_summary.csv`, `reference_profile.json`). Computed before any evaluation report existed; v1 on the 59 development scans. Medians.

	RadGenome GLI report	RadGenome impression	MultiCaRe post-treatment MRI text	v1 (development)
Words [IQR]	125 [110, 135]	42 [33, 49]	81 [47, 134]	105 [94, 122]
Sentences	4	1	4	6
Words per sentence	27.2	33.5	23.8	17.8
FKGL	16.4	21.3	15.2	10.6
Sentences with "possible" / "suggest" / "considered" / "probable, likely"	17 / 11 / 11 / 7%	59 / 40 / 37 / 24%	$\leq$ 3% each	0 / 0 / 3 / 0%
Sentences with "approximately", "about" or "estimated"	20%	1%	3%	13%

The reference reports have FINDINGS followed by a short IMPRESSION that restates the key finding with a hedge. Their hedges attach to a diagnosis ("considered", "possible", "suggests"), and post-treatment change is described against a comparison ("compared with prior", "interval", "increased", "decreased", "stable"). v1 had no impression, very few hedges and short sentences.

Design changes in v2 (developed on the 59 development scans only; frozen as v2.3). The v1 facts are kept and v2 only adds fields, all computed from the record and the fitted fold model: a risk statement that carries the 120-day horizon, so the risk sentence copies and cites one field (the horizon fix); the tier with its cut points and whether the risk is above, below or close to (within 5 points of) the training-scan average, the mean predicted risk of the fold model on its own training scans; a marker for a risk within 0.5 points of a tier cut; the out-of-fold risk of the same patient's previous scan and the change in percentage points; one fixed sentence per phase on why the phase matters; the inputs the model used and did not use (images, volumes, steroid dose, symptoms, performance status, extent of resection, perfusion and diffusion); and a note when an input was not known by the scan day. The v2 prompt keeps v1's content rules (facts only, copy numbers, cite keys, no diagnosis or RANO category, no management, volumes indicative). New rules: no verbal probability words (likely, unlikely, low, high, reassuring), so position is given only by tier and average; inputs named exactly as in the facts; complete sentences of 12 to 25 words and under 180 words in total; a fixed order in a FINDINGS section, and an IMPRESSION of at most two sentences and 45 words. From the reference, v2 copies the FINDINGS and IMPRESSION structure and the comparison wording for change; it keeps the hedge on the estimate ("the model estimates", "about"); it does not copy the diagnostic hedges, which PRISM must not use, or the high reading level. Writer, decoding and checker are as in v1 (`Qwen/Qwen3.8-27B`, greedy, seed 0, structured output), but v1 and v2 both ran on one H100 instead of four. On the development set v2.3 passed 59/59 and was complete on 59/59 (four rounds); v1 rerun there passed 59/59 and was complete on 0/59.

Supplementary Table S37. Usefulness criteria and safety checks (`prism/report/usefulness.py`). Scored automatically on the summary text and its citations, against the v2 facts of the record, for both versions.

Item	Applies when	Met when
U1 risk complete	always	the risk, both 95% bounds and "120 days" are written
U2 reference frame	always	the tier is stated and agrees with the record, or the training-scan average is given
U3 phase	always	the phase of care is named; for early post-RT, also that treatment-related change can look like progression
U4 drivers	always	at least one of the two largest contributions is named
U5 not seen	always	says that the model did not use the images, the volumes or other unseen inputs
U6 flags with reasons	a flag is active	every active flag is mentioned and the reason field of each is cited
U7 prior change	a prior scan exists	the change since the prior scan is described
U8 lookahead	an input was not known by the scan day	this is stated
S1 false reassurance	always	fails if reassuring wording ("reassuring", "low risk", "unlikely", "no evidence of", ...) appears while the risk is not in the lower third or a flag is active
S2 wording mismatch	always	fails if a verbal probability term's range excludes the point estimate (very unlikely $< 10\%$, unlikely $< 33\%$, about as likely as not 33–66%, likely $> 66\%$, very likely $> 90\%$), or tier wording disagrees with the record

Usefulness score = items met / items applicable; complete = all applicable items met. *These criteria were written with v2 in mind.* v2 was given the facts and instructions that the criteria ask for, so the criteria favour v2 by design. The U7 pattern and the part of U6 that asks for cited reasons were set by reading v2 development text. U1 to U8 are lexical: they check that an item is present, not that it is well explained.

Evaluation set and plan. The evaluation records are the 249 grade 4 scans with predictions from the out-of-fold fits of CV repeat 1 (v1 in S19 used repeat 0). The refitted fold models reproduce the stored repeat 1 predictions to 8.3×10^{-17}; 123 of the 249 scans have a prior scan. These are the same scans as in v1's evaluation, with different fold fits, so risks, intervals, tiers, conformal sets and flags differ: fresh records, not fresh patients. No v2 text was produced or read on any grade 4 record before the run. Stress set: 30 perturbed evaluation records (seed 0; 10 risks moved to another tier, 10 flags switched on, 10 switched off), with the S19 pass rule. LUMIERE could not be used, because the core model needs treatment and RT date fields that LUMIERE lacks. The plan was written on 2026-10-15 before any evaluation or stress report was generated, committed and pushed as `de9c487`, and saved with a timestamp (2026-10-16T02:59:19Z) and the sha256 of every code file. After the plan only a figure step was added; no prompt, fact, checker, criterion or rule changed.

Supplementary Table S38. Prespecified decision rule, 249 grade 4 scans, CV repeat 1 (`decision_eval.json`). v2 replaces v1 as the default summary only if every rule holds. CIs: Wilson 95% for rates; 2,000-draw patient-cluster bootstrap (seed 0) for the paired difference.

Rule	v2 result	Met
P1 faithfulness: pass rate ≥ 0.95	249/249 = 1.000 [0.985, 1.000]	yes
P2a complete rate ≥ 0.90	247/249 = 0.992 [0.971, 0.998]	yes
P2b usefulness v2 − v1, lower bound > 0	+0.325 [+0.312, +0.337]	yes
G1 forbidden content $\leq 2\%$	0/249	yes
G2 S1 or S2 $\leq 1\%$	0/249	yes
G3 no active flag omitted	0 of 262 omitted	yes

Verdict under the prespecified rule: v2 replaces v1 as the default summary. The horizon fix worked: no v2 report has an unsupported "120" (0 unsupported claims overall). On the same records v1 again fails only on this, in 11 reports, all early post-RT. v1 passes 238/249 = 0.956 [0.923, 0.975] on repeat 1 against 236/249 on repeat 0, so its prespecified target is met on one repeat and missed on the other, both times by about one report.

Supplementary Table S39. v1 and v2 on the same 249 grade 4 records (`summary_eval.csv`). Paired differences with 2,000-draw patient-cluster bootstrap CIs.

	v1	v2	v2 − v1
Pass (faithfulness)	238/249 (0.956)	249/249 (1.000)	+0.044 [+0.020, +0.072]
Complete (all applicable U items)	0/249	247/249	+0.992 [+0.979, 1.000]
Usefulness score, mean	0.674	0.999	+0.325 [+0.312, +0.337]
U1 risk, interval, horizon	249/249	249/249	
U2 reference frame	248/249	249/249	
U3 phase (and why, early post-RT)	241/249	249/249	
U4 drivers named	32/249	249/249	
U5 what the model did not see	0/249	249/249	
U6 flags with reasons (180 applicable)	180/180	178/180	
U7 change since prior (123 applicable)	105/123	123/123	
U8 lookahead stated (8 applicable)	0/8	8/8	
Forbidden content, any	0	0	0
S1 false reassurance / S2 wording mismatch	0 / 1	0 / 0	
Flags active / omitted / spurious	262 / 0 / 0	262 / 0 / 0	
Impression present	–	246/249	
Words, median [IQR]	125 [107, 135]	190 [158, 214]	+68 [+63, +72]
Sentences / words per sentence, median	6 / 21.3	11 / 17.4	
FKGL, median	11.2	9.7	−1.46 [−1.71, −1.22]
Reading time at 200 words per minute, median	38 s	57 s	
Output tokens, mean	478	784	

v1's single S2 hit is tier wording that disagrees with the record. The two incomplete v2 reports name all three active flags in one sentence without their reasons, so U6 is not met; the checker still passes them, because the flags are mentioned and cited. Three v2 reports have no IMPRESSION sentence. By phase (`by_phase.csv`), v2 passes in every phase and is complete in 98% of early post-RT scans and 100% of the others; all 11 v1 failures are early post-RT. Against the reference corpora, v2 is longer

than a RadGenome report (190 against 125 words, about 1 minute to read), with a lower reading level (FKGL 9.7 against 16.4) and shorter sentences (17 against 27 words), and it has the reference's FINDINGS and IMPRESSION structure. v2 is about 50% longer than v1 and uses 64% more output tokens.

Supplementary Table S40. Stress test, 30 perturbed records (`stress.csv`). Pass = passes the checks against the perturbed record and the report follows the change.

Version	Risk moved to another tier	Flag switched on	Flag switched off	Total
v1	9/10	8/10	10/10	27/30
v2	7/10	10/10	10/10	27/30

v1's three failures are all the horizon omission. v2 passes the checks in 30/30; its three failures are all on the prespecified rule that the original risk must be absent from the text. In each case v2 states the new risk, interval and tier correctly, and the patient's risk at the prior scan, which v2 must report, rounds to the same percent as the original risk (43 and 43, 39 and 39, 11 and 11; checked against `cv_predictions_grade4.csv.gz`). In 23 of the 123 evaluation scans with a prior scan, the prior risk rounds to the same percent as the current one. *EXPLORATORY:* v2 followed the changed field in all 30; the rule cannot tell a reported prior risk from a leaked original value.

EXPLORATORY: LLM judge. `google/gemma-4-31B-it` (revision `842da379`, Apache-2.0, a different model family from the writer) rated each report 1 to 5 on six criteria, with one sentence of reason each, and compared v1 and v2 for the same scan in both orders. It saw the facts and the summary text, not the version label, citations or section keys; v2's FINDINGS and IMPRESSION headings stayed visible, so form was not blinded. No decision depends on the judge. It is an AI model rating text written by an AI model, which is not clinical evidence.

Supplementary Table S41. EXPLORATORY: mean LLM judge ratings, 249 reports per version (`judge_summary_eval.csv`; scale 1 to 5).

Version	Accuracy	Completeness	Clarity	Calibrated language	Scope	Overall usefulness
v1	4.97	2.94	5.00	5.00	5.00	3.71
v2	5.00	5.00	5.00	5.00	5.00	5.00

Overall usefulness v2 − v1: +1.29 [+1.21, +1.38] (computed after the run, not part of the decision). Pairwise, v2 was preferred in both orders for 152/249 scans and v1 for 5/249; for 92/249 the preference flipped with order (the judge preferred whichever text came first). **The judge is not reliable.** It gave 5 on clarity, calibrated language and scope to every report of both versions. It gave v2 5 on completeness in 248/249, including the two reports that omit the flag reasons, with the wrong reason "all required elements ... are present". Its 6 accuracy deductions for v1 are its own errors: it called "upper third, values above 0.5" wrong when the cut point is 50%, which is correct. It mostly

restates the completeness checklist that the v2 prompt was built to satisfy, and adds little beyond the automatic criteria.

EXPLORATORY: further observations (an AI assistant reading the outputs, not a clinician).

- Numbers written as words escape the numeric checker: 16/249 v2 reports, for example “scan number one”, “four days before diagnosis”, “at least one input”. The five day counts and the “scan number one” cases that were checked match their records. v1 had none. The checker should map number words to numbers before it is relied on further.
- “a 11%” and “a 8%” in 15 v2 reports, from the prompt’s template “The model estimates a” followed by the risk statement.
- Raw input names copied with their value, for example “MGMT status unknown: yes” or “within 84 days after RT end: no” (170 v2 reports contain “: yes” or “: no”).
- The impression sometimes drops the tier cut points or the average value, a consequence of the 45-word limit.

These are style items for a later prompt version, which would need a fresh evaluation; nothing was changed here. The checker is lexical and does not see meaning errors, number words or grammar. Greedy vLLM batching is not bit identical across runs.

S21.1 Protocol for a future clinician reader study

Aim. To find out whether the PRISM summary helps clinicians interpret the risk number for a post-treatment glioblastoma MRI, comparing the probability alone, the probability with the v1 summary and the probability with the v2 summary.

Design and arms. Within reader, randomised and blinded. Each reader sees each case once under one of three conditions: (A) the deterministic sections only (risk, interval, tier, flags, inputs); (B) A with the v1 summary; (C) A with the v2 summary. Cases are assigned to conditions in a balanced Latin square, so each case appears under every condition across readers. A washout of at least 2 weeks applies if a reader sees the same patient twice.

Readers. At least 9 (target 12) neuro-oncologists, neuroradiologists and radiation oncologists, at least 2 of each, from at least 2 institutions, none involved in developing PRISM.

Cases. 60 grade 4 records stratified by phase of care (20 early post-RT, 20 later post-RT, 20 pre-RT or during RT) and by tier, with at least 15 with an active flag and at least 15 with a prior scan. They are drawn with a fixed seed from records not used for prompt development. No outcome is shown to readers.

Reader tasks per case (timed electronic form). (1) Comprehension, 5 items scored against the record: risk and interval; horizon; whether the risk is above or below the training average; which flag is raised and why; one input the model did not use. (2) Perceived usefulness on 1 to 7 Likert scales: "This report helps me interpret the risk estimate", "I would want this report alongside the number", "The report is clear". (3) Trust: confidence in the estimate (1 to 7) and "Is anything in the report wrong or misleading?" (free text, then adjudicated). (4) Errors: any factual error, overstatement, diagnosis or management advice. (5) Time to complete the case.

Endpoints and analysis. Primary: comprehension accuracy (items correct out of 5), C against A, in a mixed effects logistic model with random intercepts for reader and case. Secondary: B against A and C against B; usefulness scores (cumulative link mixed model); reading time; errors flagged by readers per report, with adjudicated severity; agreement of the readers with the automatic checker and with U1 to U8; and agreement of the readers with the LLM judge, to see whether the judge can be used at all. Two senior clinicians, blinded to condition, adjudicate flagged errors against the record and resolve disagreements by consensus.

Sample size. Each reader sees 20 cases per condition, so 12 readers and 60 cases give 240 reader and case pairs per condition. Detecting a 15-point difference in comprehension accuracy (0.70 to 0.85) at two-sided $\alpha = 0.05$ with 80% power needs about 120 independent pairs per condition; with an assumed design effect of 2 for clustering by reader and case this becomes about 240, that is 12 readers. The comparison within readers will likely need fewer. This rough calculation is to be redone after a pilot with 2 readers and 10 cases.

Ethics, data and reporting. Deidentified records only; images are not needed, because the report does not read them. Institutional approval and reader consent; data stay on the institutional cluster. Reporting follows the applicable CLAIM and TRIPOD+AI items, and the protocol is preregistered before the first reader session.

Supplementary figures

- **Figure S1.** Cohort flow (`docs/manuscript/cohort_flow.png`).
- **Figure S2.** Split-variance histograms per model, original analysis figure (`outputs/results/analyses/split_variance/split_variance.png`; pre-clip radiomics arm — superseded by main Figure 2).
- **Figure S3.** Phase-ladder overview (`outputs/results/analyses/phase_ladder/ladder.png`).
- **Figure S4.** Sensitivity forest plot (`outputs/results/analyses/sensitivity/forest.png`).
- **Figure S5.** Calibration curves and decision curves (`outputs/results/analyses/utility/calibration_dca.png`).

- **Figure S6.** Imaging diagnostics: label–date alignment and horizon curve (`outputs/results/analyses/diagnostics_imaging/{alignment,horizon_curve}.png`); visual QA panels (`qa/*.png`).
- **Figure S7.** SHAP summaries for the baseline and full models (`outputs/results/analyses/shap/grade4/*_{bar,beeswarm}.png`).
- [PENDING: export to the journal format.]

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